

Evolution of lateralized gustation in nematodes

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Mackie and colleagues present a **valuable** comparison of lateralized gustation in two well-studied nematodes. Their results present **convincing** evidence that ASEL/R lateralization exists and is achieved by different means in *P. pacificus* compared to *C. elegans*. This work will be of interest to neurobiologists interested in how small nervous systems make sense of the environment, and how evolution can take multiple paths to asymmetry within a neuron class.

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Abstract

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in *Pristionchus pacificus* respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode *Caenorhabditis elegans*. By visualizing neuronal activity patterns, we show that contrary to previous expectations based on its genome sequence, the salt responses of *P. pacificus* are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Animals with small nervous systems have evolved left and right asymmetry in their neurons to process various salts at different concentrations.

Introduction

Nematodes form a vast array of ecological relationships, from specialized parasite-host dependencies to nematode-microbial interactions, each one demanding exquisitely fine-tuned sets of sensory palates that span multiple modalities (1–8). Yet, the number of sensory neurons across diverse nematode species seems to be constrained (9–11). Several free-living and parasitic nematode species examined by serial section electron microscopy have nearly identical numbers of 12–13 pairs of head sensory neurons, known as the amphid neurons (11–15). How does sensory diversity arise within this constraint in neuron number? When coupled with well-described neuronal anatomy, this conserved neuron count allows for detailed comparisons at the single-cell resolution and represents an opportunity to interrogate how sensory cues are processed by anatomically similar nervous systems to produce species-specific or developmental stage-dependent behavioral outputs. To identify the genetic basis for patterning different nervous systems and to understand the processes that underlie evolutionary changes in adapting to different environments, several comparative model systems have been developed to promote comparisons to the well-studied nematode *Caenorhabditis elegans* at the genetic and cellular levels, including the predatory entomophilic nematode, *Pristionchus pacificus* (11, 16–19).

As expected from their association with insects in the wild, the olfactory preferences of *P. pacificus* are distinct from those of *C. elegans* and the human parasite *Strongyloides stercoralis* (2, 3, 20), but little is known about *P. pacificus* responses to water-soluble compounds. In *C. elegans*, the main salt receptor neuron class comprises of a bilateral pair of left and right ASE neurons (ASEL and ASER), which serve to induce an attractive locomotory response toward an increase in salt concentration (21). The gene *che-1* (chemosensory defective) encodes a transcription factor that is exclusively expressed in the ASE neurons and is required for their proper differentiation, such that a *che-1* mutant results in defective salt attraction (22–25).

One major role of *C. elegans* CHE-1 is to promote lateral asymmetry in the ASE neurons. The left and right ASE neurons asymmetrically express receptor-type guanylyl cyclases (rGCs, encoded by *gcy* genes) (26). This finding led to the realization that the ASE neurons are lateralized, such that the left and the right ASE neurons differentially respond to distinct salt ions (27–30). This observation led in turn to the identification of a complex gene regulatory network that genetically programs the distinct sensory potentials of the left and right ASE neurons (31), which includes a miRNA, *lsey-6*, at the top of this gene regulatory network (31–33). However, the *lsey-6* miRNA evolved selectively in the *Caenorhabditis* genus (34) and is absent in *P. pacificus*. Moreover, *P. pacificus* does not show an expansion of the ASEL-type and ASER-type rGCs, as *C. elegans* does (11). With these two genomic observations in mind, we had previously proposed that the ASE neurons are unlikely to be lateralized in *P. pacificus* (11).

However, we now revise this view in light of our work on mapping chemosensory responses in *P. pacificus* on the level of behavior and neuronal activity. We demonstrate that *P. pacificus* does in fact show lateralized chemosensory profiles, indicating that *P. pacificus* must have evolved independent means to establish ASE laterality. We also show that the tastant palate of *P. pacificus* is distinct from that of *C. elegans*, and that its dependence on ASE, as well as its terminal selector transcription factor *che-1*, have also diverged.

Results

P. pacificus and *C. elegans* differ in their behavioral responses to salts

Previous cross-species comparisons between *P. pacificus* and *C. elegans* indicated strong differences in olfaction preferences that reflect their divergent evolutionary histories and ecology (2024). To identify the neurons that mediate gustation, we first compared the chemosensory profiles of these two species toward water-soluble ions. In this survey, we found ammonium salts to be the strongest attractants to wildtype *P. pacificus* J4 to adult hermaphrodites (NH₄Br, NH₄Cl, NH₄I), with NH₄I significantly more attractive to *P. pacificus* compared to *C. elegans* (Fig. 1). Notably, in contrast to *C. elegans*, we find that *P. pacificus* is less attracted to NaCl and LiCl compared to the ammonium salts (Fig. 1). Also, *P. pacificus* is repulsed by acetate salts (NaAc, NH₄Ac), which induce attractive responses in *C. elegans*. We conclude that *P. pacificus* and *C. elegans* display differences in their salt preferences.

Ppa-che-1 shows similarities and difference to *Cel-che-1* in both expression and function

The *C. elegans* *che-1* mutant (*chemotaxis-defective*) was originally isolated for its inability to respond to a broad panel of salt tastants, including those described above. *che-1* was found to encode for a Zn finger transcription factor that is exclusively expressed in the ASE neuron pair, which through laser ablations had been found to be the main salt receptor neurons. *che-1* was found to control the entire differentiation of the ASE neurons, including the expression of putative receptors of the GCY receptor guanylyl cyclase family.

To assess whether *che-1* performs a similar function in salt perception for *P. pacificus* as for *C. elegans*, we analyzed the expression and function of the *Ppa-che-1* ortholog. In a previous paper, we reported that the 5' region of the sole *Pristionchus* *che-1* ortholog directs reporter expression to the *Ppa* ASE and *Ppa* ASG neuron classes. Re-examination of our provisional cell identifications using a newly-generated *che-1p::GFP* strain with stronger neurite expression revealed highly elaborated finger-like dendritic endings in the more anterior amphid neuron that could unambiguously be assigned to the AFD neurons, prompting us to reassign expression of *che-1* to ASE and AFD.

We confirmed that the *che-1* reporter transgene indicates the full expression of the endogenous *che-1* locus by tagging the endogenous *che-1* locus with an ALFA-tag. The *che-1::ALFA* animals showed staining in 2 pairs of head neurons whose position is consistent with being the ASE and AFD neurons. By crossing the *che-1* reporter transgene into a *che-1* mutant background (see below), we also found that *che-1* autoregulates its own expression especially in the ASE neurons. GFP expression in 97% in WT (n = 36) vs 4% in *che-1(-)* (n = 48), as it does in *C. elegans*. The reduction of *che-1p::GFP* in the AFDs was also observed (GFP expression in 100% in WT (n = 36) vs 33% in *che-1(-)* (n = 48)).

To provide further evidence that *Ppa-CHE-1* is indeed expressed in the *Ppa*-AFD neurons, we analyzed the expression of the *Ppa-ttx-1p::RFP* reporter. In *C. elegans*, the OTX homeodomain transcription factor TTX-1 is a terminal selector expressed in the AFD neurons required for designating the AFD fate (Satterley 2001). We found *ttx-1p::GFP* to be strongly expressed in a pair of neurons with the hallmark finger-like dendritic ending of the AFD neurons, as well as expression in other head and tail neurons (possibly RIP and RIB) and cells likely to be the

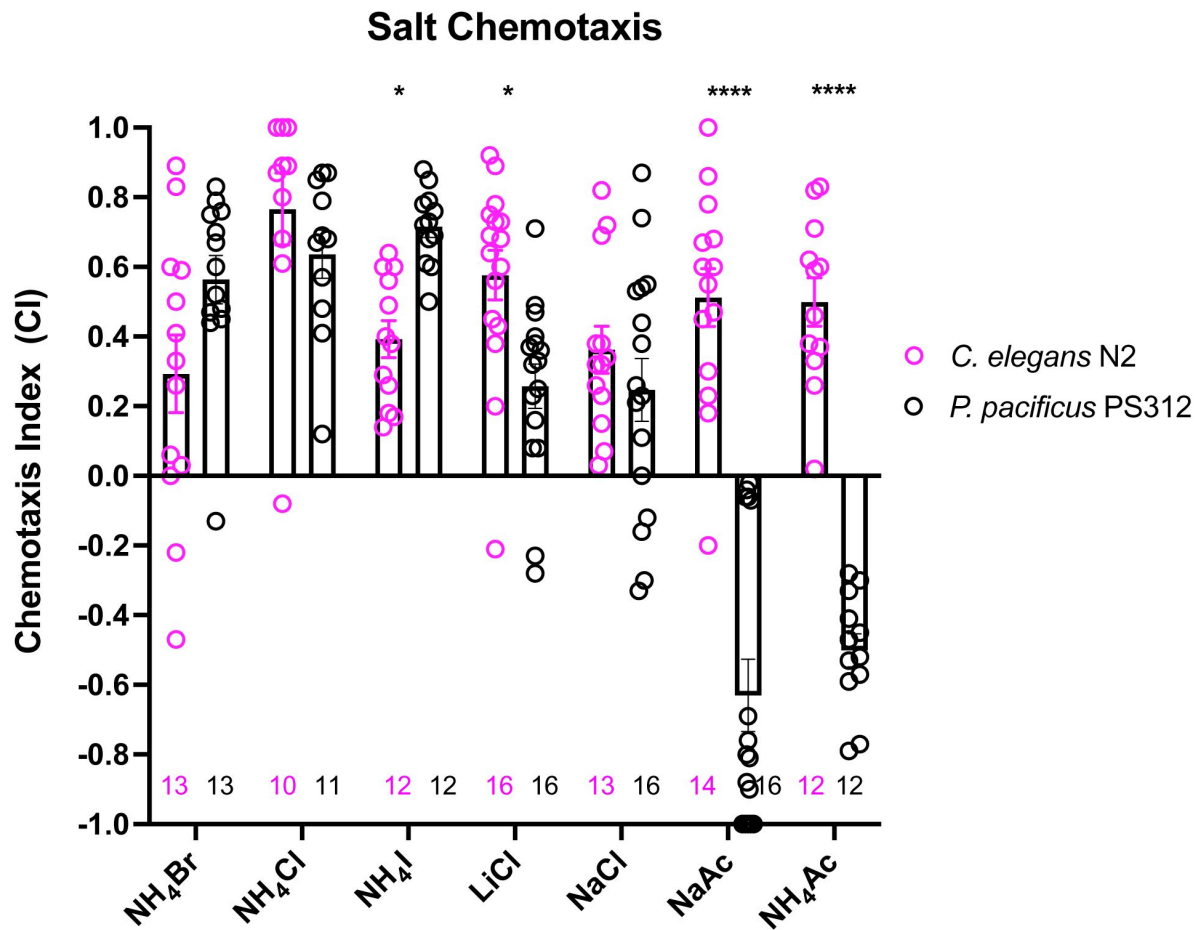


Figure 1.

A comparison of chemotaxis responses to water-soluble ions between *P. pacificus* and *C. elegans*.

J4 to Adult hermaphrodites from the two species responded to NH₄I, LiCl, and acetates significantly differently. Using two-way ANOVA, significant difference found between wildtype *P. pacificus* and *C. elegans* for the same salt is indicated above each pair (*p<0.05, ****p<0.0001), while the differences within *P. pacificus* is as follows: all salts showed difference when compared to NaAc and to NH₄Ac (****p<0.0001), but not between NH₄Ac and NaAc. Both LiCl and NaCl attraction are significantly lower than NH₄Cl (*p<0.05) and NH₄I (***p<0.001). Error bars denote standard error of the mean and the sample sizes are indicated on the bottom.

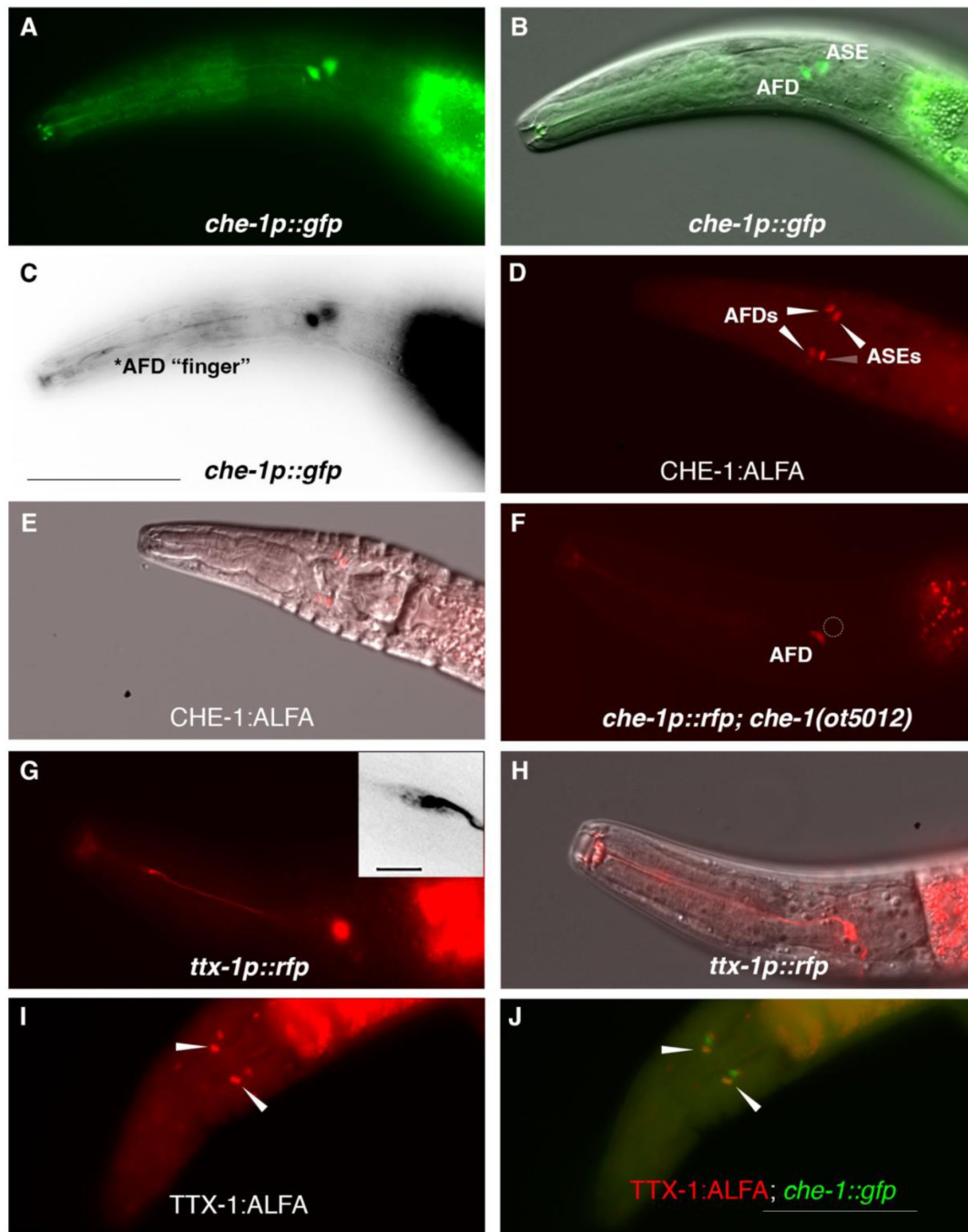


Figure 2.

***P. pacificus che-1* expression in ASE and AFD amphid neurons.**

(A-B) The *che-1::GFP* marker in the *che-1::RCaMP* reporter strain is expressed in the ASE and AFD amphid neurons. (C) *che-1::GFP* expression is detectable in the morphologically distinct AFD neurons with 'finger'-like dendritic endings. (D-E) Immunostaining of *CHE-1:ALFA* shows two pairs of amphid neuron cell bodies corresponding to the ASE and AFD neurons; dorsal-ventral view (n=114). (F) The loss of *che-1* results in loss *che-1::RFP* expression in the ASE (circle) while retaining reduced AFD expression. (G) *ttx-1p::RFP* expression in the AFD neurons with 'finger'-like dendritic endings. Inset shows expanded inverted black-white image of the AFD dendritic ending. (H) AFD expression of the same *ttx-1::RFP* animal shown in (G) in a different plane with cell body in focus. (I-J) Immunostaining of *TTX-1:ALFA* (red) shows one pair of amphid neuron cell bodies co-localizing with the anterior pair of *che-1::GFP*-expressing AFD neurons (yellow); dorsal-ventral view (n=13). Anterior is left and the scale bar in (C) represents 50 μ m for all panels except for the G inset, which is 5 μ m.

pharyngeal marginal cells based on likely conservation with *Cel-ttx-1* expression (Reilly 2022). To confirm that endogenous TTX-1 protein expression is co-expressed with *che-1p::GFP*, we examined C-terminally tagged *ttx-1::ALFA* animals and found co-localization in the same anterior pair of amphid neurons, but no co-expression in the posterior pair of amphid neurons (**Fig. 2I-J** [4](#), n=15). Interestingly, the promoter expression in the posterior pair of amphid neurons in animals with both *ttx-1p::RFP* and *gcy-22.3p::GFP* reporters do co-localize in the ASER neurons (**Fig. S1** [4](#)), possibly due to differences between the expression patterns of the 4 possible *ttx-1* splice forms. The transcriptional and protein co-expression of *ttx-1* and *che-1* in the AFD neurons unequivocally show the expression of *che-1* in both AFD and ASE neurons in *P. pacificus*.

Next, we examined whether *P. pacificus* salt responsiveness shows similar *che-1* dependence as in *C. elegans*. We generated two putative null alleles in the *Ppa-che-1* homolog using CRISPR/Cas9 genome engineering, through introduction of small deletions in the first exon of the gene, thereby resulting in frameshift and premature stops (**Fig. 3A** [4](#)). Both *Ppa-che-1* alleles exhibited defects in attraction towards NH_4Cl and LiCl compared to wildtype (**Fig. 3B** [4](#)). However, unlike in *C. elegans*, *Ppa-che-1* mutants showed no detectable difference in responses to NH_4I , NaCl , and NaAc .

We considered two different possibilities for the behavioral differences of *Ppa-che-1* and *Cel-che-1* mutants. *P. pacificus* may use sensory neurons other than ASE to sense these cues, or alternatively, *Ppa-che-1* may not have the same fundamental impact on ASE function in *P. pacificus* as it does in *C. elegans*. To explore these different possibilities, we silenced *che-1* expressing neurons by expressing codon-optimized HisCl1 channel under control of the *che-1* promoter. Histamine-treated *che-1p::HisCl1* animals showed complete loss of attraction to NH_4Br , NH_4Cl , and NH_4I but did not significantly alter their repulsive response to NaAc (**Fig. 3C** [4](#)). As a control, we show that the presence of the *che-1p::HisCl1* transgene was necessary for the knockdown of NH_4Br attraction (**Fig. S2** [4](#)). These findings corroborate that NaAc is sensed by neurons other than ASE (or AFD, in which the *che-1* promoter also drives HisCl1). Since NH_4I sensation is affected by silencing of *che-1(+)* neurons but is unaffected in *che-1* mutants, ASE function may be more greatly impacted by the silencing of ASE than by the loss of *che-1*.

The ASE neurons show left/right asymmetric responses to salt

To assess whether *Ppa* ASE neurons show the same lateralized response to salt ions as *Cel* ASE neurons, we generated transgenic *P. pacificus* lines that express RCaMP in ASE neurons and assessed calcium responses to attractive salts (**Fig. 4** [4](#), S3-S4). Specifically, we looked for changes in calcium levels immediately after the addition and removal of specific salts. When 250 mM NH_4Cl is administered, an 'ON' response is observed as calcium transiently increases in the left ASE neuron (ASEL). In contrast, an 'OFF' response was observed as calcium sharply dips in the right ASE neuron (ASER) before quickly returning to baseline when the salt was removed, which suggests hyperpolarization of this neuron (**Fig. 4A-B** [4](#))(29 [4](#)). However, when presented with a 10-fold lower concentration of 25 mM NH_4Cl , the 'OFF' response completely disappeared in ASER while the 'ON' response became more pronounced in ASEL. The ASER responses to 250 mM and 25 mM NaCl (**Fig. 4D** [4](#)) were very similar to the 'OFF' response (including hyperpolarization) observed for NH_4Cl , but the ASEL responses differ between the two salts: instead of the 'ON' response expected of ASEL, we observed a relatively weak 'OFF' response without the characteristic hyperpolarization but accompanied by an attenuated 'bump', a profile that we classify as an 'OFF-2' response (**Fig. 4C** [4](#)). Finally, we examined the response to NH_4I and found that it also elicited laterally asymmetric responses, but yet again distinct from the responses to both NH_4Cl and NaCl , with ASEL showing a strong 'ON' response, and ASER showing an 'ON-OFF' biphasic response to 250 mM NH_4I (**Fig. 4E-F** [4](#))(40 [4](#), 41 [4](#)). Interestingly, the ASER exhibited an 'OFF'-only response to 25 mM NH_4I , which was not observed for the same concentration of NH_4Cl and NaCl , and thus may reflect the higher response to NH_4I than to NH_4Cl and NaCl in the behavior assays. Altogether, *P. pacificus* ASE neurons clearly show left/right asymmetric responses to salt attractants and these asymmetric responses show similarities and differences to the *C. elegans* ASE taste neurons (see Discussion).

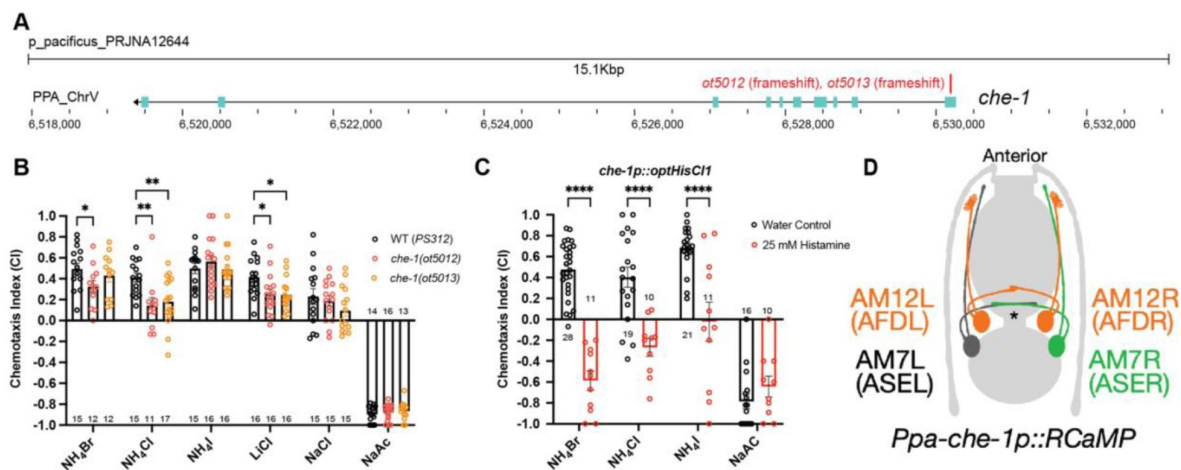


Figure 3.

***che-1* expressing amphid neurons are required for sensing water-soluble ions in *P. pacificus*.**

(A) The *che-1* locus with CRISPR/Cas9-induced mutations in Exon 1. (B) The *che-1* mutants show defects in attraction towards NH₄Br, NH₄Cl, and LiCl. Sample sizes are indicated below for attractants and above for repellent. (C) The *che-1p::HisCl1* animals lose attraction towards NH₄Br, NH₄Cl, and NH₄I in a histamine dependent manner. Sample sizes are indicated at the base of each bar. (D) A schematic map of the *P. pacificus* AM7(ASE) and AM12(AFD) amphid neurons that express the *che-1p::RCaMP* used in calcium imaging. The ASE axons are the only amphid axons in *P. pacificus* to cross each other over the dorsal lateral midline contralaterally. **P<0.01, *P<0.05 Two-way ANOVA with Dunnett's post-hoc comparison showing alleles with significant difference to wildtype *P. pacificus* (PS312). ****P<0.0001 Two-way ANOVA showing significant difference between water control and histamine treatment.

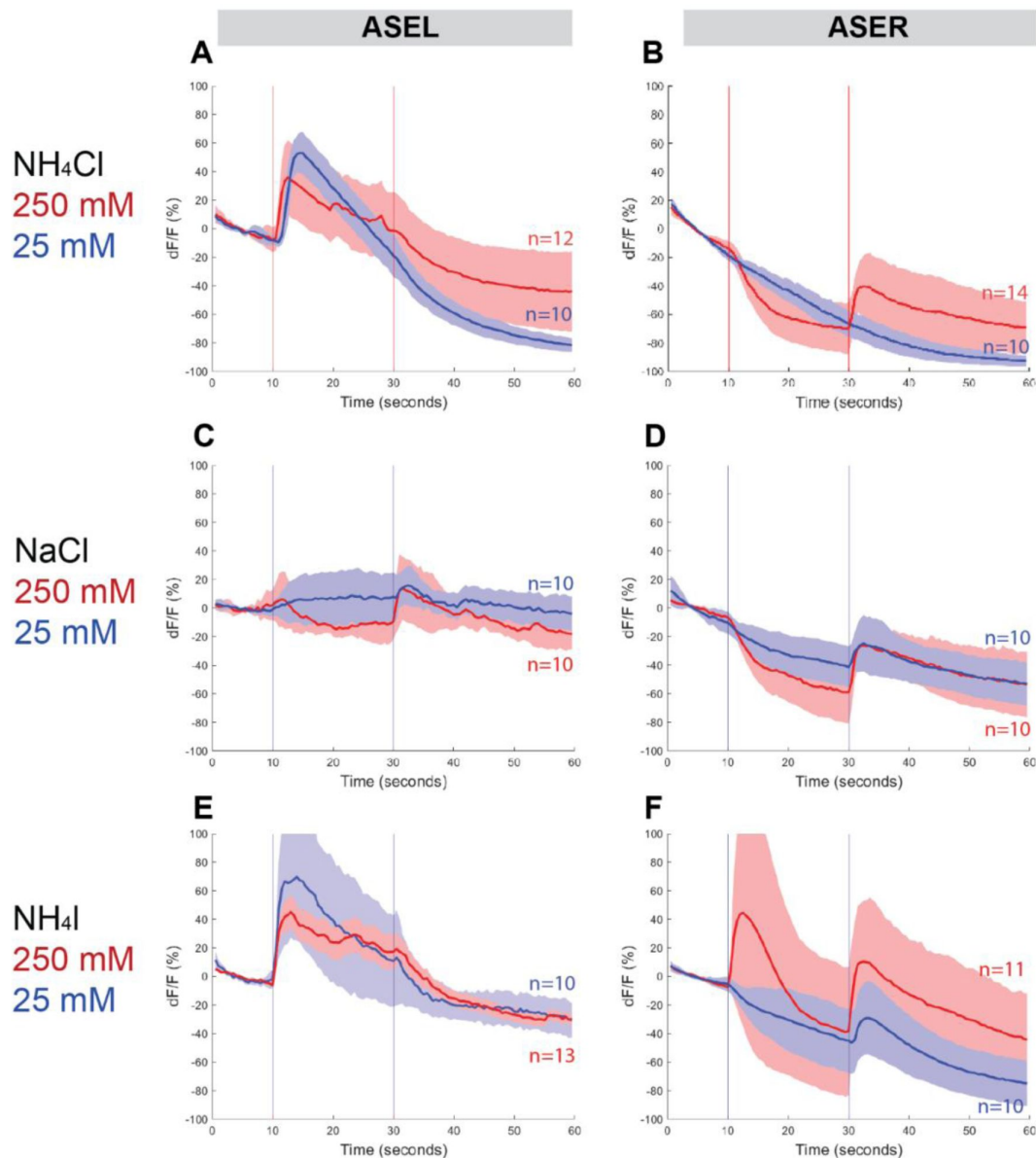


Figure 4.

ASEL and ASER responses to different concentrations of NH_4Cl , NaCl , and NH_4I .

Average percent change in RCaMP fluorescence (dF/F) over time (seconds) of tracked ASE (left and right) sensory neurons in *P. pacificus*. Salts were presented at 10s ("ON", left vertical line) for a duration of 20s, and then removed ("OFF", right vertical line) for the remaining 30s; the total recording time was 60s. (A-B) ASEL and ASER neuron responses to high (250 mM, red) compared to low (25 mM, blue) concentrations of NH_4Cl . (C-D) ASEL and ASER neuron responses to NaCl . (E-F) ASEL and ASER neuron responses to NH_4I . Shaded ribbons represent 95% confidence intervals. Shaded ribbons in (E-F) have been cropped to maintain consistent y-axes across the plots, allowing for easier comparison. Sample sizes are indicated (n).

The AFD neurons also respond to salts in *P. pacificus*

The RCaMP line that we used to assess calcium responses in ASE is also expressed in AFD, allowing us to simultaneously examine the calcium responses in the AFD (AM12) neurons (Fig. 5 [↗](#), S1 [↗](#), S5 [↗](#)-S7 [↗](#)). Surprisingly, we detected a distinctly ‘ON-OFF’ biphasic response to all 3 salt types at both concentrations. Specifically, although we observed weaker or comparable responses in AFD neurons when compared to either ASE neuron’s response towards 250 mM NH_4Cl , NaCl and NH_4I (Fig. 5A-B [↗](#), 5E-F, 5I-J), the AFD responses were more robust than ASER toward 25 mM NH_4Cl , NaCl, and NH_4I (Fig. 5D [↗](#), 5H [↗](#), 5L [↗](#)). Specifically, AFD neurons responded strongly to 25 mM NaCl when neither one of the ASE neurons showed a positive response (Fig. 5G-H [↗](#)). Averaging the calcium transients separately by AFD left versus right did not result in significant differences in the shape of the neuronal calcium responses, with the exception of the AFDR responses to higher versus lower concentrations of NH_4I (Fig. S6 [↗](#)-S7 [↗](#)). We have not further pursued whether these AFD responses are a reflection of a direct perception of salt or a secondary consequence of communication of salt-perceptive neurons (like ASE) to AFD.

A target of *che-1*, the guanylyl cyclase *gcy-22.3* is required for chemotaxis

We further explored the asymmetric salt perception by the ASE neurons, which in *C. elegans* is largely mediated through distinct receptor-type guanylyl cyclases (rGC proteins, encoded by *gcy* genes), which confer salt specificity via their extracellular domains (30 [↗](#)). Our previous genome survey of *Ppa* homologs of *gcy* genes has revealed patterns that made us question whether *Ppa* *gcy* genes are convincing candidates for lateralized chemotactic responses. Specifically, we noted that ASER-expressed *C. elegans* *gcy* genes and ASEL-expressed *C. elegans* *gcy* genes have only expanded in the *Caenorhabditis* genus (11 [↗](#)). One outlier to this pattern is the *Cel-gcy-22* gene, which is expressed in ASER, but has not expanded in *C. elegans* (11 [↗](#), 42 [↗](#)). However, this gene has duplicated several times in *P. pacificus*, resulting in 5 putative *Ppa-gcy-22* paralogs (11 [↗](#)). We fused the promoter of one of these paralogs, *Ppa-gcy-22.3*, to *gfp* and found that transgenic animals express GFP exclusively in ASER (Fig. 6A [↗](#)), identical to the *C. elegans* *gcy-22* ortholog (42). We confirmed its expression in ASER by analyzing animals that carry both the *Ppa-gcy-22p::gfp* reporter and the *Ppa-che-1p::rfp* reporter, showing a unilateral overlap of these reporters in ASER (Fig. 6B [↗](#)).

To assess whether *Ppa-gcy-22.3* is a potential effector of *Ppa-che-1* function, we crossed the *gcy-22.3* reporter into *che-1(ot5012)* mutants. We found expression of *gcy-22.3* was eliminated (Fig. 6C [↗](#)), leading us to conclude that *gcy-22.3* is a potential effector of *che-1* function, identical to its homolog in *C. elegans*.

To determine whether and which of the observed salt responses is mediated by the ASER-expressing *gcy-22.3*, we generated a putative *gcy-22.3* null mutant through CRISPR/Cas9 genome editing (2 bp complex deletion that introduces a frameshift)(Fig. 7A [↗](#)) and examined its response to the higher salt concentration. The ‘OFF’ response to 250 mM NH_4Cl was notably abolished in the ASER neuron in the loss-of-function *gcy-22.3* mutant, while the ‘ON’ response in the ASEL remained intact (Fig. 7B-C [↗](#)). However, the responses to 250 mM NaCl were not significantly reduced in either the ASEL or ASER neuron in the *gcy-22.3* mutant (Fig. 7D-E [↗](#)). Furthermore, the *gcy-22.3* mutation also reduced the ‘ON’ portion of the AFD biphasic response following presentation of 250 mM NH_4Cl and NaCl (Fig. 7F-G [↗](#)). We also examined the behavioral responses toward individual salt ions in *gcy-22.3* mutants, including a second loss-of-function allele *csu182*. Overall, we found no significant differences between wildtype and mutants in responses towards individual ions, and only the attraction to NH_4Cl was slightly enhanced in both alleles (Fig. 8 [↗](#)), which was unexpected given the lack of calcium response observed in *gcy-22.3(cs181)* towards

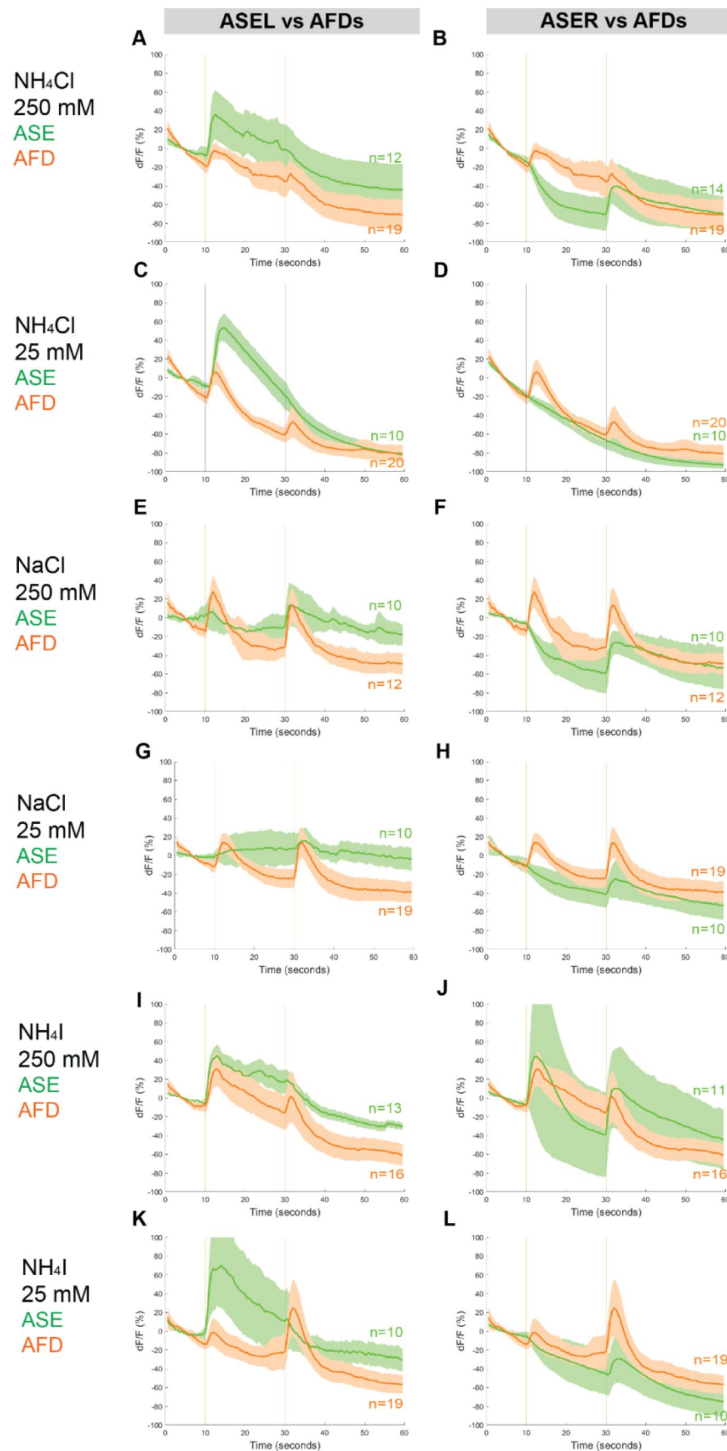


Figure 5.

Combined AFD responses in comparison to ASE left or right neuron responses to NH_4Cl , NaCl , and NH_4I .

Average percent change in RCaMP fluorescence (dF/F) over time (seconds) as described in [Fig. 4](#). Averaged combined AFD (left and right neurons, orange) compared to left or right ASE (green) responses to (A-B) 250mM NH_4Cl (C-D) 25mM NH_4Cl (E-F) 250mM NaCl (G-H) 25mM NaCl (I-J) 250mM NH_4I (K-L) 25mM NH_4I . Shaded ribbons represent 95% confidence intervals. Shaded ribbons in (J-K) have been cropped to maintain consistent y-axes across the plots, allowing for easier comparison. Sample sizes are indicated (n).

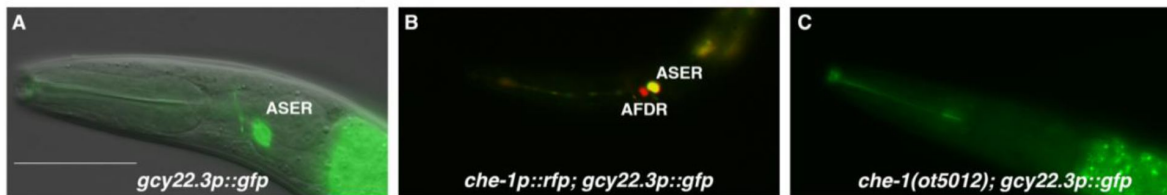


Figure 6.

The laterally asymmetric expression of *gcy-22.3* is dependent on the zinc finger transcription factor CHE-1.

(A) The *gcy-22.3::GFP* marker is expressed exclusively in the right ASE neuron (ASER)(n>200). (B) The *gcy-22.3::GFP* marker co-localizes with *che-1::RFP* expression in the ASER. (C) *gcy-22.3::GFP* expression is absent in the *che-1(ot5012)* mutant (n=55). Anterior is left and dorsal is top. Scale bar in (A) represents 50 μm for all panels.

NH₄Cl. Our findings show that while proper ASE function is critical for salt attraction, defects in individual *gcy* genes do not lead to a major impact on the worms' ability to track toward attractive salts.

Discussion

Our study has revealed several insights into the substrates of evolutionary changes between two distantly related nematode species, *P. pacificus* and *C. elegans*. We used intracellular calcium levels as our readout for neuronal activity with a genetically encoded calcium sensor in two pairs of *che-1*-expressing amphid sensory neurons– the first calcium imaging study in *P. pacificus*. We show that three neuron types (ASE left, ASE right and AFD neurons) each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. We have identified the first laterally asymmetric marker in *P. pacificus*, *gcy-22.3p::GFP*, with its expression limited to ASER homolog (AM7). Our unexpected discovery that neuronal asymmetry is present in the ASE homologs between two distantly related nematode species, despite the lack of a *lsy-6* homolog, suggests that functional lateralization in *P. pacificus* may be mediated by a different genetic pathway compared to *C. elegans*.

P. pacificus and *C. elegans* have diverged taste palates

We have shown that while *C. elegans* is attracted to acetate salts, *P. pacificus* avoids these acetates. Previous studies have shown that ammonium acetate (NH₄Ac) is sensed both as a water-soluble compound as well as a volatile odorant and is mediated by different signaling pathways (36). Like *C. elegans*, it is therefore likely that ammonium and acetate ions involve a different set of neurons in *P. pacificus* (i.e. non-*che-1* expressing neurons), based on the finding that histamine-treated *che-1p::HisCl1* animals did not significantly attenuate their repulsive response to NaAc. We find that sodium chloride is a common attractant for both *P. pacificus* and *C. elegans*, although the magnitude of response is lower than previously published results (29). This result is likely due to differences in generating the salt gradients. Nevertheless, we find that *P. pacificus* neurons have a distinct response to this salt when compared to those observed in *C. elegans* neurons. The *P. pacificus* ASEL neuron responds to a decrease in NaCl concentration (likely sodium), as evidenced by the 'OFF-2' response profile, whereas the *C. elegans* ASEL neuron responds to an increase in sodium concentration. However, in both nematodes, the ASER neuron responds to a decrease in chloride concentration. Additionally, the *P. pacificus* ASER neuron exhibits a unique ON-OFF response to ammonium iodide, whereas in *C. elegans*, no ON-OFF type of response is seen in the ASE neurons– the *C. elegans* ASER neuron responds only to a decrease in iodide concentration. Based on our calcium imaging and chemotaxis results on the *gcy-22.3* mutants, attraction to ammonium ions (NH₄⁺) is likely mediated in part by the ASER neuron in *P. pacificus*. *Pristionchus* species are entomophilic and most frequently found to be associated with beetles in a necromenic manner and thus insect cadavers could be sources of ammonium in the soil (6, 43–45). Additionally, ammonium salts could represent a biological signature of other nematodes that the predatory morph of *P. pacificus* could interpret as prey. In *P. pacificus*, nutritional state has a measurable role in the mouth-form polyphenism decision between predatory and non-predatory morphs (46). Collectively, our findings highlight the divergence of the *P. pacificus* salt-sensory neurons compared to those observed in *C. elegans*.

P. pacificus ASE neurons have narrower sensitivity range

The sensitivity range of the *P. pacificus* ASER responses is significantly less compared to the ASEL responses, as well as to *C. elegans* ASER responses. Whereas the *C. elegans* ASER has a 40-fold sensitivity range in the 'OFF' response to the removal of various concentrations of NaCl (1–40 mM) (29, 47–49), the *P. pacificus* ASER showed the 'OFF' response only to 250 mM NH₄Cl but not to a 10-fold reduction in concentration of NH₄Cl (25 mM). For the ASEL in contrast, the response to

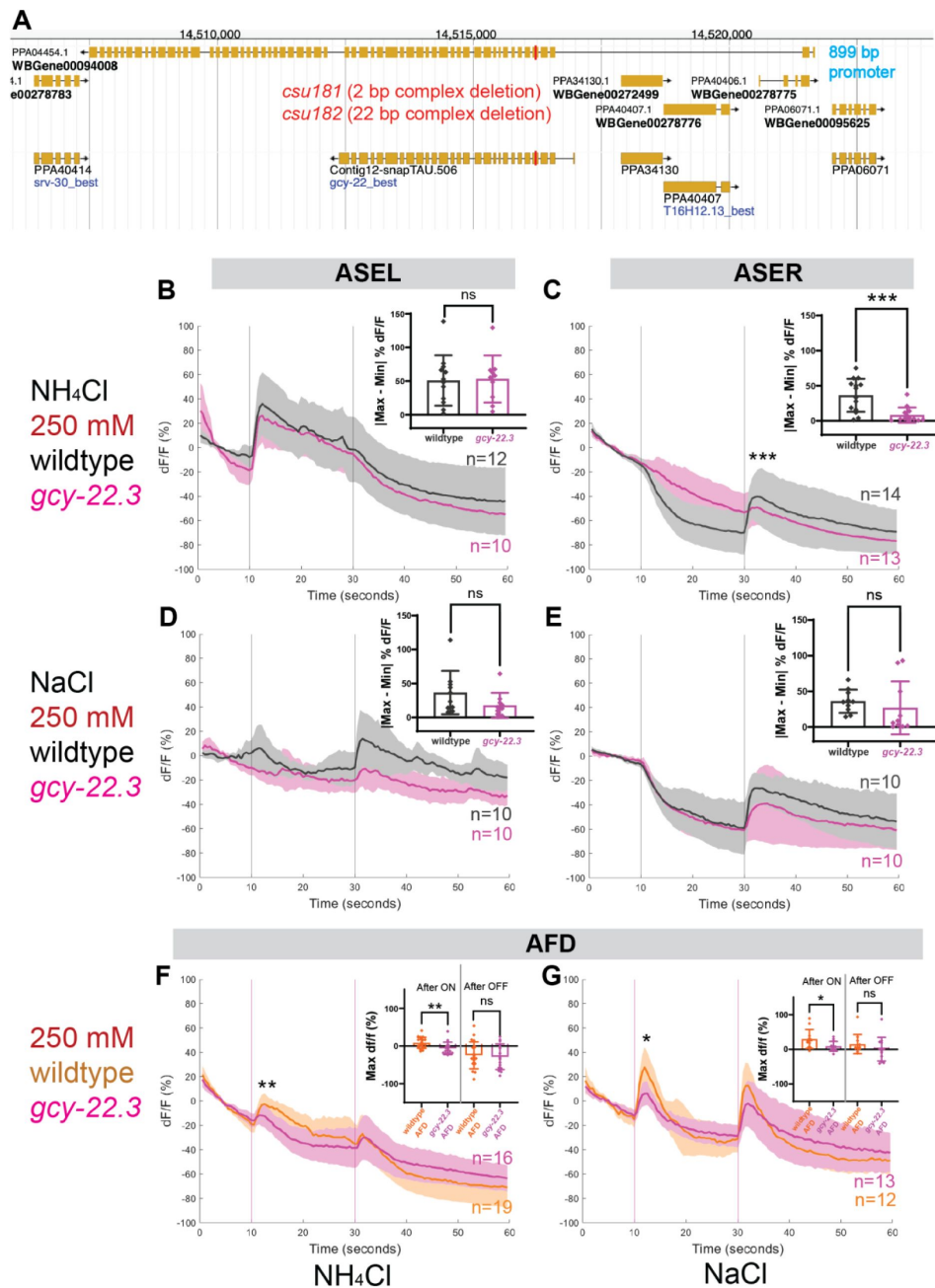


Figure 7.

ASE and AFD responses in wildtype compared to *gcy-22.3* mutants.

Average percent change in RCaMP fluorescence (dF/F) over time (seconds) as described in Fig. 4 (A) www.pristionchus.org Genome Browser view of the *gcy-22.3* locus with the CRISPR/Cas9-induced mutations indicated with a red bar (Exon 5 of PPA04454 or Exon 4 of Contig12-snapTAU.506). (B-C) ASEL and ASER neuron responses to 250 mM NH_4Cl wildtype (gray) and *gcy-22.3* mutants (magenta). (D-E) ASEL and ASER neuron responses to 250 mM NaCl in wildtype (gray) and *gcy-22.3* mutants (magenta). AFD (combined left and right neurons) responses (F) to 250 mM NH_4Cl and (G) to 250 mM NaCl in wildtype (orange) and *gcy-22.3* mutants (magenta). Shaded ribbons represent 95% confidence intervals. Sample sizes are indicated (n). For ASEL/R comparisons, bar plots represent the difference between the minimum % dF/F value 10s pre-stimulus and maximum % dF/F value 10s post-stimulus for either the (B) ON or (C-E) OFF stimulus. For AFD comparisons, bar plots represent maximum % dF/F values 10s after the ON and OFF stimulus. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ indicate significant difference. "ns" indicate no significant difference. Comparisons between different genotypes were analyzed using unpaired t-test or Mann-Whitney test.

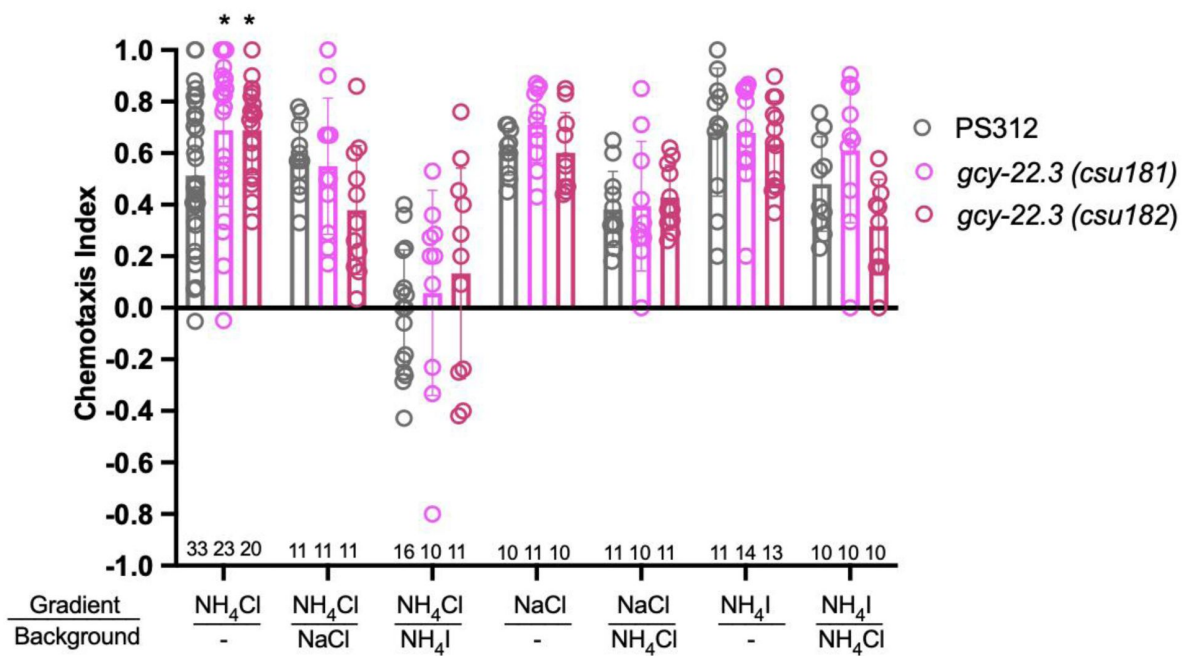


Figure 8.

Chemotaxis responses to individual ions in *gcy-22.3* mutants.

Young adult hermaphrodite responses to salt gradients with or without various salts in background. *P<0.05 Significant differences were found between wildtype PS312 and *gcy-22.3* mutants to NH₄Cl by two-way ANOVA and Dunnett's test. Each assay involves a minimum of 10 animals. Sample sizes for each condition are indicated on the bottom. Error bars denote standard error of the mean.

25 mM was just as strong as to 250 mM NH_4Cl (10-fold) and comparable to the 8-fold concentration range observed for *C. elegans* ASEL towards NaCl. Alternatively, the magnitude of these sensitivity differences may be also partially due to differences among calcium indicators (*i.e.* GCaMP and Cameleon), but multiple *P. pacificus* *che-1p::GCaMP* strains did not exhibit sufficient basal fluorescence to allow for image tracking and direct comparison. The narrower sensitivity range in *P. pacificus* chemosensation was also observed for attraction to volatile odors (21), which span only 10-fold, versus up to 10,000-fold in attractive odors for *C. elegans* (21).

***P. pacificus* taste neurons exhibit a unique biphasic response**

Although the left-ON and right-OFF responses are conserved in the ASE neurons in both species, the biphasic response of ASER to 250 mM NH_4I has not been observed in either of the ASE neurons towards salts. In *C. elegans*, hyperosmotic stimulus such as 1M glycerol, or high concentrations or long duration of CuSO_4 exposure both result in biphasic responses by the ASH neurons that sense noxious chemicals (41, 50). Olfactory neurons that mediate avoidance behavior such as the AWB neurons can also exhibit a biphasic response to the presence and removal of a high concentration isoamyl alcohol that normally elicits avoidance behavior (51). The neuron-specific response can also be dependent on the concentration of the chemical compound, since the ASER to 25 mM NH_4I was a weak ‘OFF’ rather than a biphasic one. In contrast to the salt-dependent response types by *P. pacificus* ASE neurons, the *P. pacificus* AFD neurons also show exclusively biphasic responses with various amplitudes. Further characterization will help determine if biphasic responses can also be found in other sensory neuron types in *P. pacificus*, specifically those neurons mediating avoidance behavior.

AFD are potentially polymodal neurons

Broadly speaking, *C. elegans* chemosensory neurons have been classically characterized as specialized neurons for dedicated modalities such as water-soluble chemicals (ASE), volatile odorants (AWA, AWB, AWC), noxious chemicals (ASH), pheromones (ADL, ADF, ASK), as well as light (ASJ) and temperature (AFD) (1, 37, 52–55). Advances in multi-neuron calcium recordings have since shown that a given odor within a certain concentration range is detected by different ensembles of the 12 amphid neuron classes, including the AFD neurons (56–58). Unexpectedly, we found that the *P. pacificus* AFD neurons exhibit a distinctive biphasic response to all three salts tested (NH_4Cl , NaCl, and NH_4I), which differ from the *C. elegans* AFD calcium responses (58, 59). Moreover, the loss of the receptor *gcy-22.3* reduced the AFD ‘ON’ response to NH_4Cl and NaCl, indicating ASER contributes to the AFD response. The strong positive AFD response to 25 mM NaCl that is absent in both ASE neurons further supports the likelihood that AFD receives inputs from other amphid neurons. The integration of thermosensation and chemosensation is important for memory-regulated behavior. In *C. elegans*, maximum chemotaxis indices toward NH_4Cl occurs when there is concordance between cultivation temperature and assay temperature (60). The *C. elegans* AFD neurons are also important for gustatory aversive learning in NaCl avoidance (49). Given the influence of environmental temperature on the *P. pacificus* mouth-form plasticity and the wide range of micro-climates that wild strains of *P. pacificus* have been isolated from (61–63), temperature and taste preferences could be regulated at multiple genetic levels during crucial developmental decisions.

Changes in the gene regulatory architecture of sensory neuron specification

We found that the key regulator of *Cel* ASE identity, *che-1*, is also expressed in *Ppa* ASE and may play a similar role as a terminal selector in this neuron type, based on its effect on ASE-mediated behavior and regulation of the *Ppa-gcy-22.3* gene. However, the stronger behavioral effect of silencing of *che-1* expressing neurons compared to a *che-1* mutant background could either indicate that *che-1* does not have as broad a role in controlling ASE differentiation in *P. pacificus* versus *C. elegans*. It is possible that a developmental loss of ASE differentiation may result in

compensatory changes in the chemosensory system during early development, as was observed in the *C. elegans* mating pheromone response by males (64). It is also worthwhile to note that *Cel-gcy-22* stands out for being located on a separate chromosome (Chr. V) compared to the other ASER-type *gcy* genes (*gcy-1*, *gcy-4*, *gcy-5* on Chr. II) as well as the *Cel-gcy-22* mutant having defects in chemoattraction towards a wide range of salt ions (30, 42). Given that the 5 *P. pacificus* *gcy-22*-like paralogs are located on 3 separate chromosomes (Chr. I, IV, X), it is likely that they emerged from independent and repeated gene duplication events after the separation of *Caenorhabditis* and *Pristionchus* lineages. Although the promoter fusion reporters of other *gcy* genes have been uninformative due to lack of expression (*Ppa-gcy-22.1*, *Ppa-gcy-7.1*, *Ppa-gcy-7.2*, *Ppa-gcy-5*), it is likely that ASEL-specific *gcy* genes as well as additional ASER-specific *gcy-22*-like genes exist. Finding other genes with left-right specific expression could help to identify genetic determinants affecting lateral asymmetry.

Unexpectedly, we found that unlike in *C. elegans*, the *Ppa-che-1* gene is also expressed in the AFD neurons. Since we cannot record neural activity in AFD in a *che-1* mutant (the *che-1p::RCaMP* driver fails to be expressed sufficiently in *che-1* mutants due to autoregulation), and do not yet have molecular markers for *Ppa* AFD neurons, we cannot assess whether *che-1* affects AFD neuron differentiation.

Perhaps the most striking difference in the gene regulatory architecture of ASE neuron specification is the apparent lack of the key regulator of ASE asymmetry in *P. pacificus*, the miRNA *lsy-6*. In *C. elegans*, the expression of *lsy-6* exclusively in ASEL is prepatterned via an early embryonic Notch signal (33) and serves to downregulate the homeodomain transcription factor *cog-1* in the ASEL neuron (65). Through a network of downstream regulatory events, asymmetry of rGCs eventually becomes established (31). *cog-1* and several asymmetrically expressed downstream effectors of *cog-1*, such as the *die-1*, *lim-6* and *fozi-1* transcription factors are conserved in *P. pacificus*, but whether the function of these factors in controlling *P. pacificus* ASE laterality is conserved remains to be determined. In this context it is intriguing to note that another prominent gene regulatory pathway that is controlled by miRNAs in *C. elegans*, the heterochronic pathway (*let-7s*, *lin-4*), appears to have diverged in *P. pacificus* as well, despite the conservation of the overall physiological readouts of this pathway (temporal patterning of cell lineage divisions)(66). It is tempting to speculate that miRNA-mediated regulatory process is particularly labile.

In conclusion, our work illustrates how comparative behavioral and genetic analyses in nematodes is a powerful strategy to uncover substrates of evolutionary change in simple nervous systems.

Materials and Methods

Nematode Strains

P. pacificus and other nematode strains were maintained at ~20°C on NGM plates seeded with *E. coli* OP50 for food as described previously (67); these are derived from standard *C. elegans* culture methods (68). *P. pacificus* and other nematode strains used are listed in **Table S1**.

Chemotaxis assays

The assay for assessing response to salt gradients was adapted from *C. elegans* and *P. pacificus* chemotaxis assays (2, 21, 42). Overnight salt gradients were established on 10 cm chemotaxis plates containing 20 ml agar (5 mM KPO₄, 1 M CaCl₂, 3% Bacto-agar, 1 mM MgSO₄) by adding 10 µl of 2.5 M salt solutions for 16 hours. Alternatively, agar containing 25 mM (NH₄Cl, NH₄I) or 50 mM (NaCl) were used to test for responses to individual salt ions. Following the establishment of the overnight point gradient, another 4 µl of the same salt solution or water

control was added to reinforce the gradient 4 hours before the assay. Just prior to the assay, 1 μ l of 1 M sodium azide was added to both the attractive salt (A) and the control (C) spots. *P. pacificus* J4 to adult hermaphrodites from near-saturated cultures were washed 3x with distilled water and collected by centrifuging at 2000 rpm for 2 minutes. Approximately 200 worms were loaded onto the edge of each assay plate between the gradient sources, and at least 10 combined worms have to reach the scoring arenas to be considered. At least 10 assays constituted each experimental trial, and multiple trials were conducted and averaged for each condition. The Chemotaxis Index (CI) for each end-point assay plate is defined as $(A - C)/(A + C)$. To conduct conditional knockdowns of neurons, 5 M histamine dihydrochloride (Sigma Aldrich H7250) stock solution in sterilized deionized water (Arrowhead CA) was filter-sterilized and top plated onto the agar plates to a final histamine concentration of 25 mM histamine approximately 10 minutes before loading the worms to commence the assay. Most assays lasted 3-4 hours at room temperature to allow *P. pacificus* sufficient time to reach the scoring arenas, with ~40% of the animals participating. When using the *csuEx83[Ppa-che-1p::optHisCl1]* strain, only animals expressing *Ppa-egl-20p::RFP* tail marker from the extrachromosomal array were scored. Because of *P. pacificus*' strong aversion to acetate, we could not easily assess the individual contributions of salt ions in a saturated background of ammonium acetate as conventionally practiced in *C. elegans* studies (30).

Ppa-che-1p::HisCl1 strain

To make the *Ppa-che-1p::optHisCl1*, the *P. pacificus* codon-optimized histamine-gated chloride channel sequence used in *C. elegans* (*HisCl1*) (69, 70) was designed using (https://hallemmlab.shinyapps.io/Wild_Worm_Codon_Adapter/) (71) and custom synthesized (Twist Bioscience), and subsequently inserted behind the *che-1* promoter (3.1 kb containing the first exon and intron) (11) to create the pH30 plasmid construct. This *Ppa-che-1p::optHisCl1* plasmid (2 ng/ μ l) along with PS312 genomic DNA (80 ng/ μ l) and *Ppa-egl-20p::RFP* co-injection marker (2 ng/ μ l) were individually digested with HindIII and assembled as the injection mix to create *csuEx83*.

Ppa-che-1p::RCaMP reporter strain

To make the *Ppa-che-1pei::optRCaMP*, we generated a transgenic worm strain expressing the codon-optimized genetically-encoded calcium indicator (GECI), jRCaMP1a, in the neurons of interest (72, 73). jRCaMP1a is an improved red GECI based on mRuby with comparable sensitivity to GCaMP6 (74). The codon-optimized RCaMP sequence was custom synthesized (Twist Bioscience), and then subcloned using the plasmid pJET1.2/blunt (Thermo Fisher Scientific) to create the pMM2 plasmid construct. Using Gibson Assembly (E2611, New England Biolab), the codon-optimized RCaMP sequence was introduced downstream of the *Ppa-che-1* promoter sequence with the first exon and intron sequences to create the pMM5 plasmid construct. Separately, we also generated a *Ppa-che-1pei::optGFP* transcriptional reporter with codon-optimized GFP (pMM3) to enable the localization of the *che-1*-expressing neurons during video acquisition (75). The *Ppa-che-1pei::optRCaMP* (2 ng/ μ l) and *Ppa-che-1pei::optGFP* constructs (1 ng/ μ l), along with PS312 genomic DNA (80 ng/ μ l) and *Ppa-egl-20p::RFP* (1.5 ng/ μ l) were individually digested with HindIII and assembled as the injection mix to create *csuEx93*. Despite multiple attempts, we were unable to generate an equivalent *che-1pei::GCaMP* transgenic line with sufficient basal level of GCaMP expression for a comparison to the RCaMP calcium dynamics.

Promoter fusion reporter strains

The *Ppa-gcy-22.3p::GFP* construct contains 899 bp of the upstream promoter of the PPA04554 transcript is fused to the codon-optimized GFP (pVL2). A shorter *gcy-22.3* transcript with different first two exons is also predicted, which is more similar in length to other rGC genes in *P. pacificus* (Contig12-snapTAU.506). The *Ppa-ttx-1pei::RFP* construct contains 1935 bp of the upstream promoter along with the first exon and intron sequences of *Ppa-ttx-1* (PPA26714) fused to the codon-optimized RFP, excluding the first 2 codons (pDC14). pVL2 (2 ng/ μ l) or pDC14 (2 ng/ μ l), along with PS312 genomic DNA (80 ng/ μ l) were individually digested with HindIII to create the injection

mixes to generate the independent reporter strains, *csuEx90[Ppa-gcy-22.3p::GFP]*, as well as *Ppa-ttx-1pei::RFP(csuEx94)* and *Ppa-ttx-1pei::RFP(csuEx96)*. *csuEx96* showed less gland cell expression (which can occlude neuronal expression) and stronger AFD expression than *csuEx94*.

CRISPR mutagenesis generated mutants

CRISPR/Cas9 mutagenesis was used to generate mutations ([75](#), [76](#)). crRNA and primer sequences, and induced mutations, are included in [Table S3](#) and [Fig. S8](#).

che-1 alleles (PPA01143)

Target crRNA, tracrRNA, and Cas9 nuclease were purchased from IDT Technologies (San Diego, CA). crRNA and tracrRNA were hydrated to 100 μ M with IDT Duplex Buffer, and equal volumes of each (0.61 μ l) were combined and incubated at 95°C for 5 minutes, then 25°C for 5 minutes. Cas9 protein (0.5 μ l of 10 μ g/ μ l) was added, then the mix was incubated at 37°C for 10 minutes. *Ppa-egl-20p::RFP* was used as a co-injection marker. To reach a final total volume of 40 μ l, the Cas9-crRNA-tracrRNA complex was combined with pZH009 (*Ppa-egl-20p::RFP*) DNA to reach 50 ng/ μ l final concentration using nuclease-free water. F₁ progeny were screened for the presence of *Ppa-egl-20p::RFP* expression in the tail and candidate F₁'s were sequenced to identify heterozygotes ([76](#)). *ot5012* has a 4 bp insertion while *ot5013* has an 8 bp complex insertion/deletion, and both mutations cause frameshift mutations and premature stop codons. Each allele was outcrossed two times to wildtype before characterization.

gcy-22.3 alleles (PPA04454)

Target crRNA, tracrRNA, and Cas9 nuclease were purchased from IDT Technologies (San Diego, CA). crRNA (RHL1400) and tracrRNA were hydrated to 100 μ M with IDT Duplex Buffer, and equal volumes of each (0.61 μ l) were combined and incubated at 95°C for 5 minutes, then 25°C for 5 minutes. Cas9 protein (0.5 μ l of 10 μ g/ μ l) was added, then the mix was incubated at 37°C for 10 minutes. *Ppa-egl-20p::RFP* was used as a co-injection marker. To reach a final total volume of 40 μ l, the Cas9-crRNA-tracrRNA complex was combined with pZH009 (*Ppa-egl-20p::RFP*) DNA to reach 50 ng/ μ l final concentration using nuclease-free water. F₁ progeny were screened for the presence of *Ppa-egl-20p::RFP* expression in the tail and candidate F₁'s were sequenced to identify heterozygotes ([76](#)). *csu181* has a 2 bp complex deletion, while *csu182* has a 22 bp complex deletion, both mutations cause frameshifts and premature stop codons. Each allele was outcrossed two times to wildtype before characterization.

ALFA C-terminal tagging and immunostaining

For induction of site-specific insertions via CRISPR/Cas-9 mediated mutagenesis, target crRNA, tracrRNA, and Cas9 nuclease were treated as described above. Single-stranded DNA repair template containing the ALFA nanobody tag (RHL1551 for CHE-1 and RHL1519 for TTX-1) with 35 bp of homology arms on the 5' and 3' sides were purchased from IDT. To minimize sequence identity, the two copies of the ALFA sequence contain silent mutations. The crRNA (RHL1396 for *che-1* and RHL1514 for *ttx-1*) and tracrRNA were hydrated to 100 μ M with IDT Duplex Buffer, and equal volumes of each (0.61 μ l) were combined and incubated at 95°C for 5 minutes, then 25°C for 5 minutes. The ALFA tag was C-terminally inserted to the longest *ttx-1* splice form in the last 18th exon (*ppa_stranded_DN30925_c0_g1_i5*) (Trinity 2016 transcripts can be found on www.pristionchus.org [77](#)). F₁ animals expressing the co-injection marker *egl-20::optRFP* were lysed and checked by PCR for insertions.

Non-starved healthy cultures of ALFA-tagged CHE-1 (*csu226[Ppa-che-1::2xALFA]*) or ALFA-tagged TTX-1 (*RLH280[Ppa-ttx-1::2xALFA]*) (from six 6 cm plate cultures) were washed with M9 and filtered (Sartorius 84g/m2, Grade 392) and processed as previously described ([39](#)). In brief, mixed stage worms were fixed overnight at 4°C on a nutator with 500 μ l fixation buffer (4%

paraformaldehyde in PBS). The worms were then incubated overnight at 37°C with 500 µl 4% β-mercaptoethanol dissolved in 1% Triton-X100 in 0.1M Tris pH 7.4, and digested in 200 µl of collagenase buffer with 200 units of collagenase type IV (Gibco, Thermo Fisher Scientific, Waltham, MA, USA) at 37°C for ~3.5 hours. The partially digested worms were subsequently washed three times with 500 µl PBST. After collagenase treatment the centrifugation between the washes was done at low speed, 1000 RCF. After washes, the worms were stained in 50 µl 1% BSA in PBST with the primary antibody (1:100 FluoTag-X2 anti-ALFA-AZdye568, N1502, NanoTag Biotechnologies, Göttingen, Germany), overnight at 4°C in a nutator. Following the washes, worms were resuspended in 50 µl VectaShield mounting medium (H-1000, Vector Laboratories, USA) containing DAPI (Molecular Probes, Thermo Fisher Scientific) and gently mounted on freshly prepared 3% Noble agar pads. Images were acquired on a Leica DM6000 microscope.

Calcium Imaging

To conduct calcium imaging, worms were trapped and imaged within a microfluidic PDMS chip while delivering stimuli directly to the nose of an immobilized worm, as previously described (50, 77). The microfluidic chip was connected to a programmable valve controller (ValveBank) that enables the user to toggle between “stimulant ON” and “stimulant OFF” states. Specifically, the ValveBank allows controlled switching of flow from the two outer buffer channels in the chip such that either the control solution (stimulant OFF) or the stimulant solution (stimulant ON) flows over the worm nose (Fig. S9). Buffer solution (for the outer two channels) consisted of M9 buffer with 0.1% Tween-20 and 1 µg/ml fluorescein (78, 79). The worm loading solution consisted of M9 buffer with 0.1% Tween-20 and 1.5 mM tetramisole hydrochloride to immobilize the animal. Water-soluble stimulant solutions consisted of the following water-soluble compounds dissolved in nanopure/milli-Q water: ammonium chloride (NH₄Cl), ammonium iodide (NH₄I), and sodium chloride (NaCl) at concentrations of 250 mM and 25 mM. 750 mM NaCl elicited very inconsistent responses. The water-soluble stimulant solution was made by diluting the water-soluble compound in nanopure/milli-Q water with 0.1% Tween-20. The control solution consisted of 0.1% Tween-20 nanopure/milli-Q water. Worms were exposed to the control solution, stimulant solution, and then control solution using a 60-second program: 10 seconds stimulant OFF, 20 seconds stimulant ON, and 30 seconds stimulant OFF. As a negative control, animals were exposed only to the control solution, without switching channels, for the duration of the recording. For each animal, the orientation of the nose and vulva were recorded and used as a guide to determine the ventral and dorsal sides of the worm, and subsequently, the left and right sides of the worm. Accounting for the plane of focus of the neuron pairs as viewed through the microscope, it was then determined whether the imaged neuron was the worm’s left or right neuron of each pair. Images were captured using a Zeiss Axio Observer Z1 inverted fluorescence microscope and a pco.panda 4.2 sCMOS camera. Changes in fluorescence intensity were measured in the neurons of interest while the worm was exposed to green light. Images were processed using MetaMorph software version 7.10.5.476. Images were captured at 500 ms exposure time at 2 fps because the baseline fluorescence in *Ppa-che-1p::optRCaMP* worms was too dim to capture viable data using the standard 100 ms exposure time. Baseline F_0 was measured as the average background-subtracted fluorescence from the first 9 seconds of each recording and change in fluorescence intensity was calculated as $dF/F = (F - F_0) / F_0$, as described (79). Data were analyzed and plotted using custom scripts generated in MATLAB versions R2021a–R2024a. This code is available at <https://github.com/honglabcsun/Calcium-Imaging>.

For bar plot comparisons between wildtype and *gcy-22.3* mutants, we calculated minimum pre-stimulus and maximum post-stimulus % dF/F values using custom scripts generated in MATLAB. Min-Max data were exported as text files, manually converted and organized into an XLSX file (Microsoft Excel Office16) and imported into Prism GraphPad software (version 10) for data visualization and statistical analysis.

Nomenclature

Throughout the results section, *P. pacificus* genes will be referred to without the *Ppa*- prefix; if necessary for comparison to another species such as *C. elegans* (*Cel*-) the *Ppa*- prefix will then be used.

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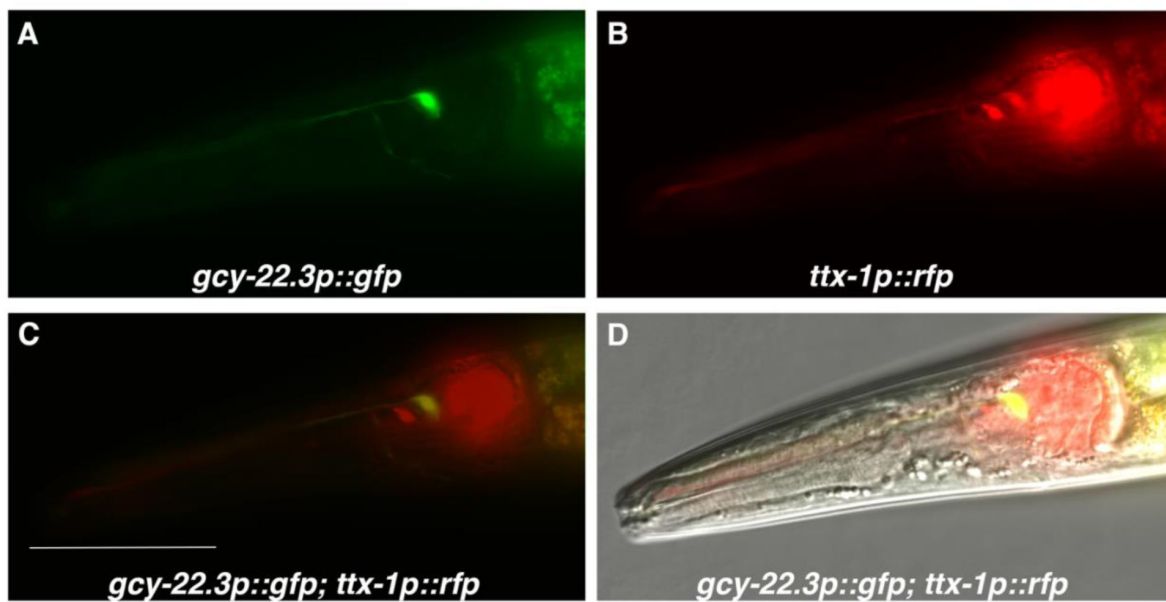


Figure S1.

Co-expression of *gcy-22.3p::GFP* and *ttx-1p::RFP* in ASER.

A representative F₁ male animal from a cross between the two reporter strains (n=5). Scale bar in (C) represents 50 μm for all panels.

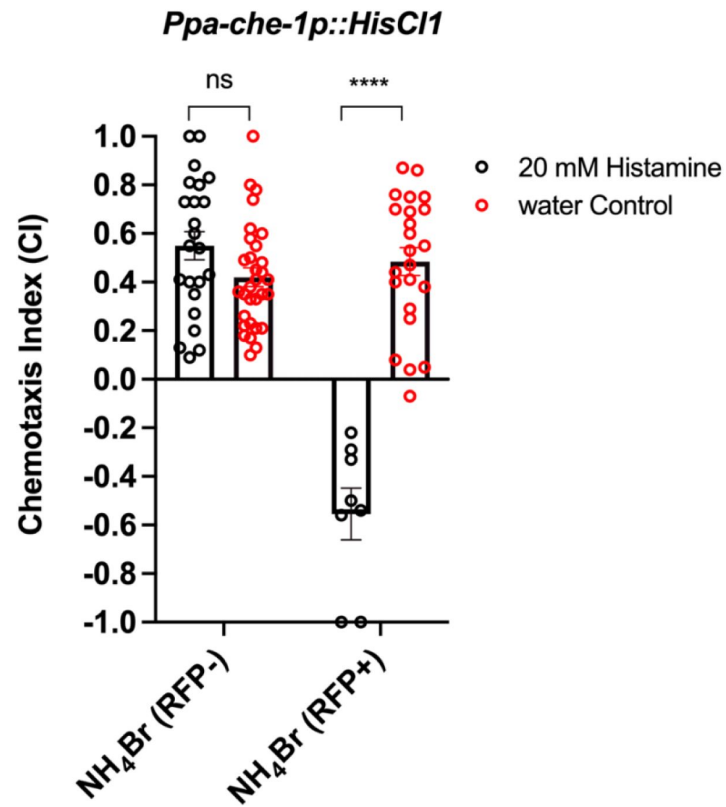


Figure S2.

The *che-1p::HisCl1* transgene is necessary for histamine-dependent knockdown of salt attraction.

Animals containing the *che-1p::HisCl1* transgene and the co-injection marker *egl-20p::RFP* were scored separately on each assay plate. Unpaired t-test between histamine and water treatment prior to chemotaxis assays (**** $P < 0.0001$).

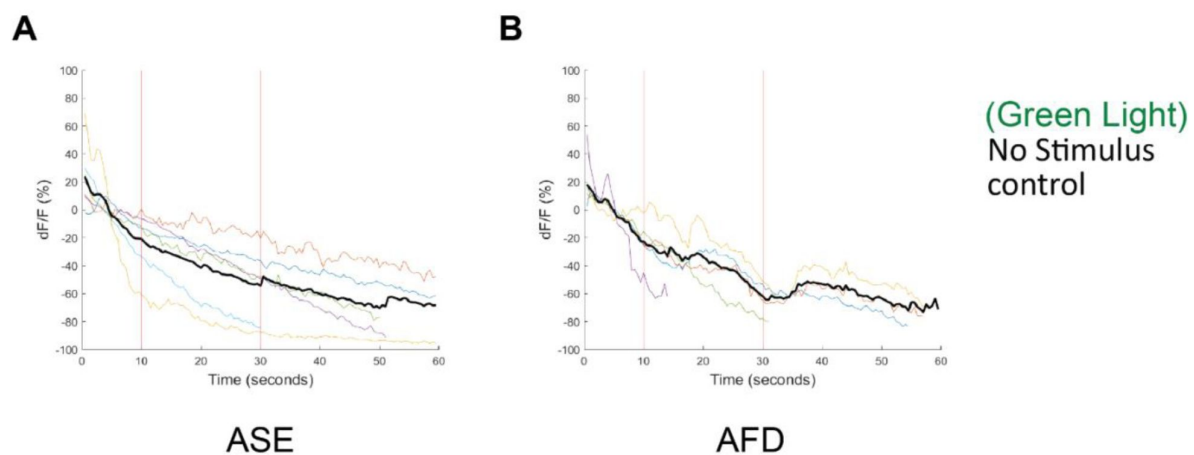


Figure S3.

Negative controls of ASE and AFD neuron responses to green light.

Average percent change in fluorescence (dF/F) over time (seconds) of tracked sensory neurons in *P. pacificus* (black line). Individual animals are represented by colored lines. No salt was presented; vertical lines indicate time points where salt was presented and removed for experimental samples, shown for comparison. *csuEx93[Ppa-che-1p::optRCaMP]* worms experienced the control solution for the duration of the recording and were imaged under green light. (A) ASE ($n=6$). (B) AFD ($n=5$).

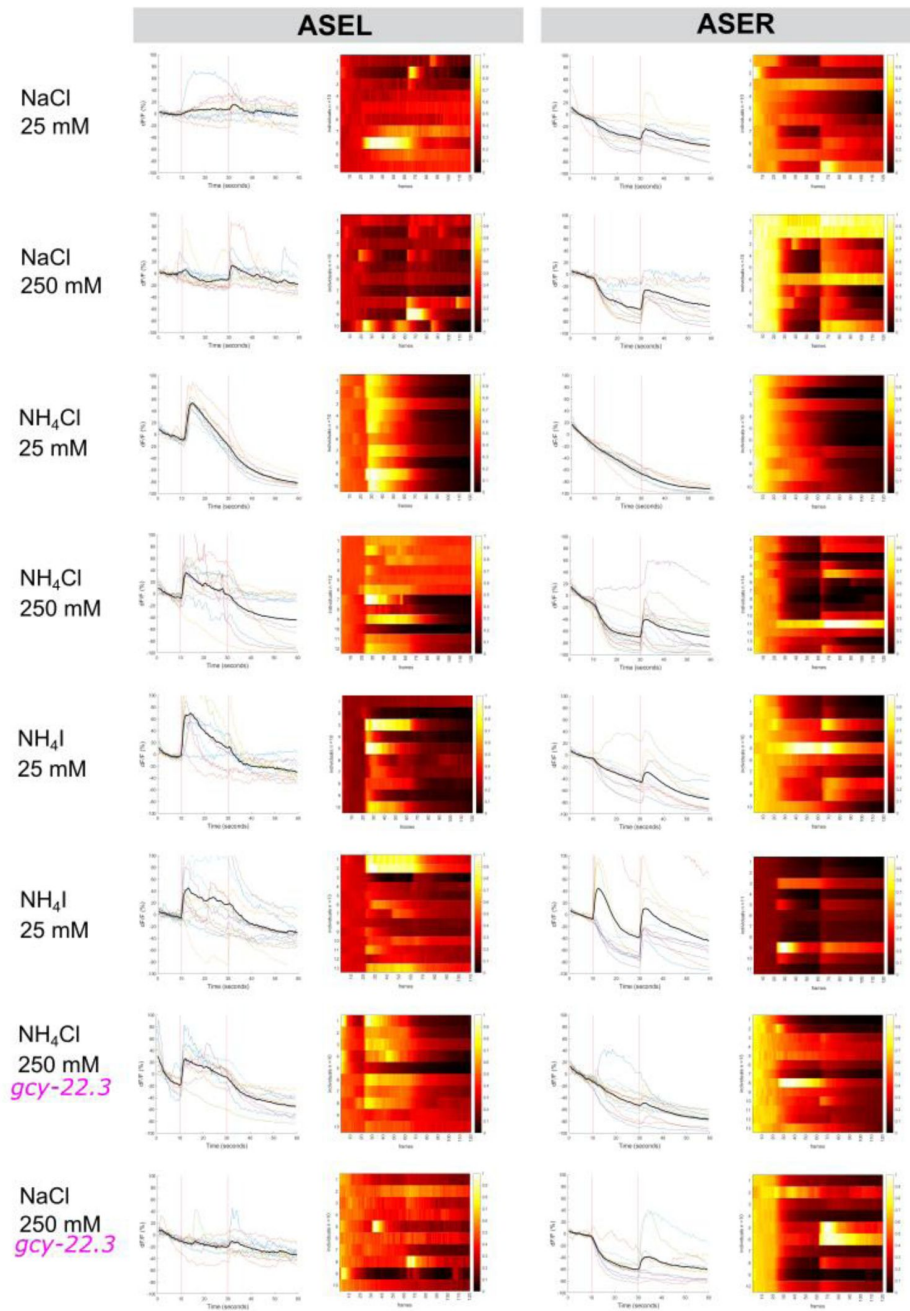


Figure S4.

Individual traces and heatmaps of ASE responses to various salts.

Average percent change in fluorescence (dF/F) over time (seconds) of tracked ASEL and ASER neurons in *P. pacificus* (black line). Individual animals are represented by colored lines. For heatmaps, each row represents a single individual. Lighter colors (closer to 1) represent more positive dF/F values and darker colors (closer to 0) represent more negative dF/F values.

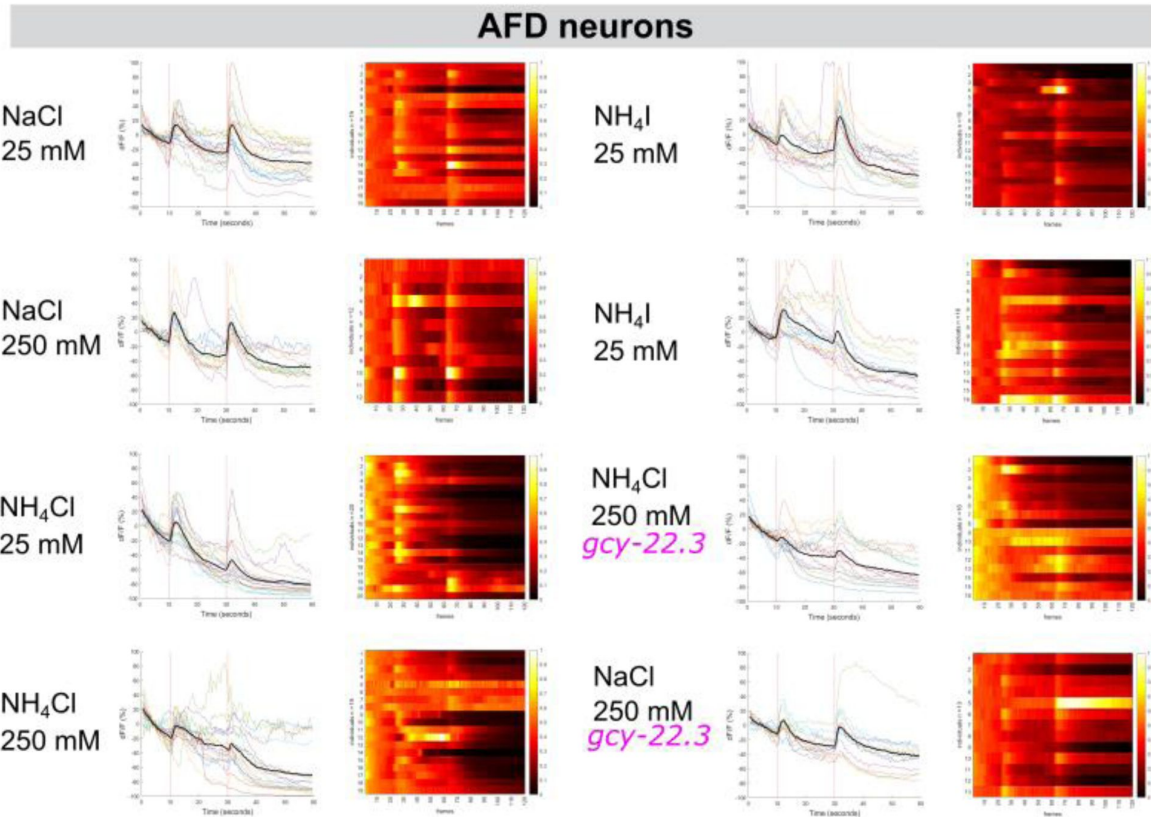


Figure S5.

Individual traces and heatmaps of AFD responses to various salts.

Average percent change in fluorescence (dF/F) over time (seconds) of tracked left and right AFD neurons in *P. pacificus* (black line). Individual animals are represented by colored lines. For heatmaps, each row represents a single individual. Lighter colors (closer to 1) represent more positive dF/F values and darker colors (closer to 0) represent more negative dF/F values.

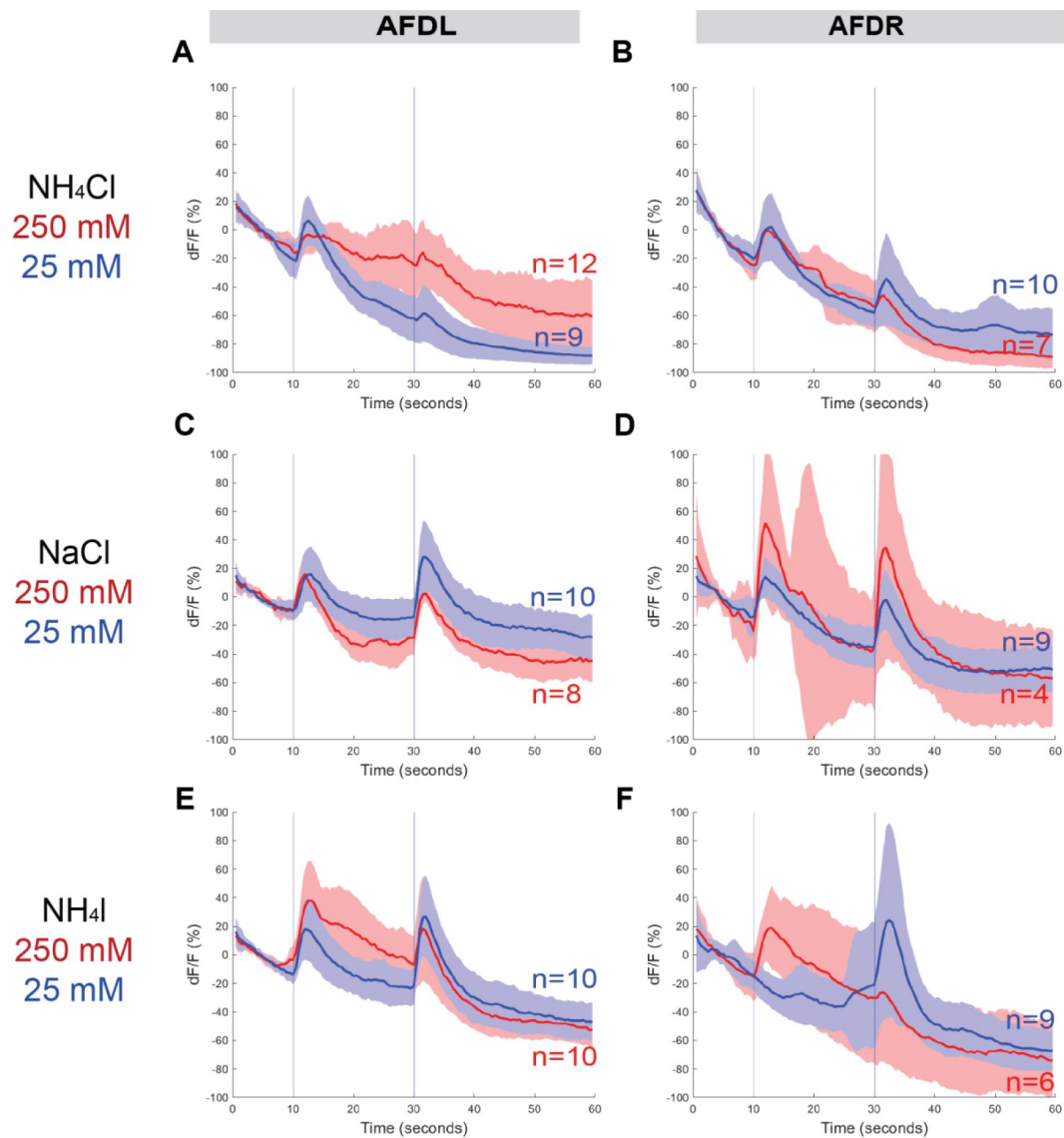


Figure S6.

AFDL/R responses to different concentrations of NH₄Cl, NaCl, and NH₄I.

Average percent change in RCaMP fluorescence (dF/F) over time (seconds) as described in [Fig. 4](#). Average AFDL and AFDR neuron responses to high (250mM, red) and low (25mM, blue) concentrations of salts: (A-B) NH₄Cl, (C-D) NaCl, (E-F) NH₄I. Shaded ribbons represent 95% confidence intervals. Shaded ribbons in (D) have been cropped to maintain consistent y-axes across the plots, allowing for easier comparison. Sample sizes are indicated (n).

Figure S7.

AFDL and AFDR responses to NH₄Cl and NaCl in wildtype and *gcy-22.3* mutant.

The combined AFD responses shown in **Figure 7F** have been separated into AFDL and AFDR. Average percent change in RCaMP fluorescence (dF/F) over time (seconds) as described in **Fig. 4**. AFDL and AFDR neuron responses to (A-B) 250mM NH₄Cl and (C-D) 250mM NaCl in wildtype (orange) compared to *gcy-22.3* mutants (magenta). Shaded ribbons represent 95% confidence intervals. Shaded ribbons in (D) have been cropped to maintain consistent y-axes across the plots. Sample sizes are indicated (n).

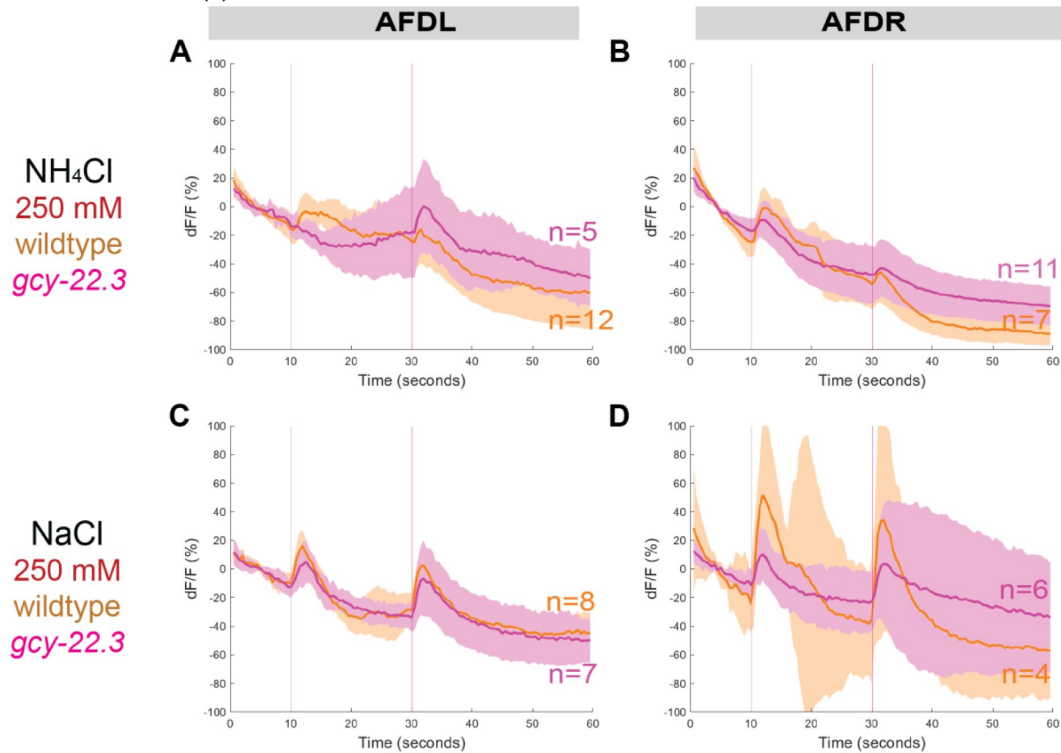


Figure S8.

Molecular lesions for *P. pacificus* *che-1* and *gcy-22.3*.

DNA alignment of the *Ppa-che-1* and *Ppa-gcy-22.3* alleles and their predicted amino acid changes.

CHE-1 (PPA01143) WT	CTACGGTCTTGGGTCGGGCTCGGCCAGAA
ot5012 (4 bp insertion)	CTACGGTCTTGGGTCCTCTGGGCTCGGCCAGAA
ot5013 (complex indel)	CTACGGTCTTGG--AACCATAGACGAACGGGTT--CCCA

CHE-1 (PPA01143) WT AA	MAYVG YD YGLG.....
ot5012 (early STOP)	MAYVG... NASDD* (frameshift +157 AA)
ot5013 (early STOP)	MAYVG... RSWTI* (frameshift +33 AA)

GCY-22.3 (PPA04454) WT	AGTGACACTGGCTCAGAAATTCGAATGGACTGATCTCACCTTTATGTTCTCTGCGGAGC
csu181 (2 bp deletion)	AGTGACACTGGCTCAGAAATTCGAATGGACTGATCTCACCTTTT TTTTGT --GCGGAGC
csu182 (complex indel)	AGTCCGTAGTAGTTAG-----CCGTAGTCCGTAGTAGTTAGC

GCY-22.3 (PPA04454) WT	PIILWGPPYQSTLEDSQAFPTMMSTSF SARPRARAVVTLAQKFEWTDLTFMFSAERDPLV...	120
csu181 (early STOP).	PIILWGPPYQSTLEDSQAFPTMMSTSF SARPRARAVVTLAQKFEWTDLTF FFCGARSARW*	120
csu182 (early STOP).	PIILWGPPYQSTLEDSQAFPTMMSTSF SARPRARAVVGPVVVSRSP*	104

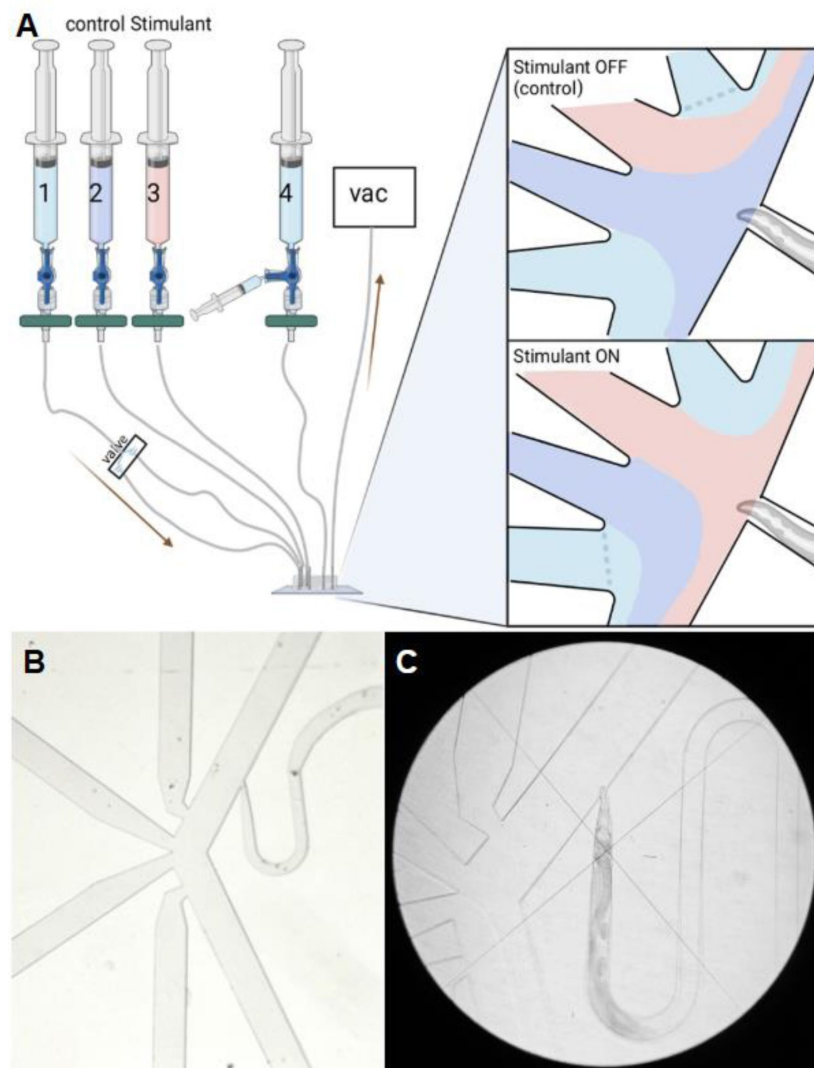


Figure S9.

(A) Schematic of the microfluidic apparatus to conduct calcium imaging while delivering stimuli directly to the nose of an immobilized worm. Syringes act as reservoirs for buffer solution (light blue), control solution (purple), and stimulant solution (red). Tubing inserts into the microfluidic PDMS chip. Panels (right) zoom in to depict a schematic of the microscope view of the PDMS chip: (top) fluid flow over the worm nose when stimulant is switched OFF (control solution flows over worm nose); (bottom) fluid flow over the worm nose when stimulant is switched ON (stimulant solution flows over worm nose). Created with *BioRender.com* <https://www.biorender.com/>. (B) Actual microscope view of PDMS chip design. The outer two channels hold buffer solution and can be switched open (ON) or closed (OFF) by the Valvebank. The inner two channels hold experimental solutions: the inner channel closer to the worm trap holds the control solution, and the inner channel farther from the worm trap holds the stimulant solution. (C) Actual microscope view of *P. pacificus* loaded into the PDMS chip while fluid is flowing. The PDMS chip features a U-shaped worm trap to facilitate loading the worm head-first, and a tapered opening to ensure the worm fits snugly and will not slide too far forward during recording.

	<i>P. pacificus</i>		<i>C. elegans</i>	
	ASEL	ASER	ASEL	ASER
NH ₄ Cl (1 – 80mM)	ON	x	x ^{29,30}	OFF ^{29,30}
NH ₄ Cl (81 – 250mM)	ON	OFF	?	?
NaCl (1 – 80mM)	x	OFF	ON ^{29,43-45}	OFF ^{29,43-45}
NaCl (81 – 250mM)	OFF-2	OFF	ON ⁴⁴	OFF ⁴⁴
NH ₄ I (1 – 80mM)	ON	OFF	X ³⁰	OFF ³⁰
NH ₄ I (81 – 250mM)	ON	ON-OFF	?	?

"x" minimal to no response

"?" represents unknown

Table 1.

Comparison of ASEL/R responses between *P. pacificus* and *C. elegans*.

Gene (<i>Ppa</i> -)	Allele	Strain	Genetic Lesion or Genotype
--	wildtype	PS312	<i>P. pacificus</i> California reference strain
<i>gcy-22.3</i>	<i>csu181</i>	RLH293	2 bp complex deletion causing frameshift and early STOP. 2x outcrossed.
<i>gcy-22.3</i>	<i>csu182</i>	RLH310	22 bp complex deletion causing frameshift and early STOP. 2x outcrossed.
<i>gcy-22.3p::GFP</i>	<i>csuEx90</i>	RLH334	<i>Ppa-gcy-22.3p::GFP</i> ; <i>Ppa-egl-20p::turboRFP</i>
<i>che-1</i>	<i>ot5012</i>	OH17879	4 bp insertion causing frameshift and STOP. 2x outcrossed.
<i>che-1</i>	<i>ot5013</i>	OH17880	8 bp complex indel causing frameshift and STOP. 2x outcrossed.
<i>che-1p::HisCl1</i>	<i>csuEx83</i>	RLH336	<i>Ppa-che-1pei::optHisCl1</i> ; <i>Ppa-egl-20p::turboRFP</i>
<i>che-1p::RFP</i>	<i>lucEx367</i>	MLC545	<i>Ppa-che-1pei::turboRFP</i> ; <i>Ppa-egl-20p::turboRFP</i>
<i>che-1::2xALFA</i>	<i>csu226</i>	RLH325	<i>Ppa-che-1::ALFA</i> (C-terminal tagged to last Exon)
<i>che-1p::RCaMP</i> ; <i>che-1p::GFP</i>	<i>csuEx93</i>	RLH335	<i>Ppa-che-1pei::optRCaMP</i> ; <i>Ppa-che-1pei::optGFP</i> ; <i>Ppa-egl-20p::turboRFP</i>
<i>che-1p::RCaMP</i> ; <i>gcy-22.3</i>	<i>csu183</i>	RLH300	<i>gcy-22.3(cs181)</i> ; <i>Ppa-che-1pei::optRCaMP</i> ; <i>Ppa-che-1pei::optGFP</i> ; <i>Ppa-egl-20p::turboRFP</i>
<i>ttx-1p::RFP</i>	<i>csuEx94</i>	RLH351	<i>Ppa-ttx-1pei::RFP</i> (promoter with first exon and intron)
<i>ttx-1p::RFP</i>	<i>csuEx96</i>	RLH352	<i>Ppa-ttx-1pei::RFP</i> (promoter with first exon and intron)
<i>ttx-1::2xALFA</i>	--	RLH280	<i>Ppa-ttx-1::ALFA</i> (C-terminal tagged to Exon 18)
--	wildtype	N2	<i>C. elegans</i> Bristol reference strain

"pei" denotes the regulatory region with promoter, exon 1, and intron 1.

Table S1.

Nematode strains.

Table S2.

Plasmids.

Plasmid	Reporter
<i>pMM3</i>	<i>Ppa-che-1pei::optGFP</i>
<i>pMM5</i>	<i>Ppa-che-1pei::optRCaMP</i>
<i>pHC30</i>	<i>Ppa-che-1pei::optHisCl1</i>
<i>pVL2</i>	<i>Ppa-gcy-22.3p::GFP</i>
<i>pDC14</i>	<i>Ppa-ttx-1pei::RFP</i>

Table S3.

Primer sequences.

Gene (<i>Ppa</i> -)	Primer	Sequence (5' → 3')
<i>che-1</i> (PPA01143)	SJC4	CTACGGTCTTGGGTCGGGCT (crRNA, exon 1)
<i>che-1</i>	SJC341	ATCTCTCTCCCTCATCTCCC (forward)
<i>che-1</i>	SJC342	TTCCTCCCTCTCCTGACAGC (reverse)
<i>che-1</i>	SJC343	AGACGGATGATTAGGGAAGC (forward, nested)
<i>che-1</i>	SJC344	GAAGGTGTGGGAGGGTTTCC (reverse, nested)
<i>gcy-22.3</i> (PPA04454)	RHL1553	TCTCACCTTTATGTTCTCTG (crRNA, exon 5)
2xALFA	ssDNA	TCTAGACTCGAGGAGGAGCTCAGAAGAAGACTCACTG AGtcCCgTctcgaggaggagctcCgTCgTCgTctcacCgag
<i>che-1</i> (C-terminal ALFA)	RHL1396	CACATGAAGACGCACAAGAA (crRNA, last exon)
<i>che-1</i> (C-terminal ALFA)	RHL1397	AGGTCACAAGCCTTATTCGTGC (forward)
<i>che-1</i> (C-terminal ALFA)	RHL1399	GAATCTGTTGCAGCATCTGTTGC (reverse)
<i>gcy-22.3</i>	RHL1401	CGGTCCGGCATGCAGTCAAC (forward)
<i>gcy-22.3</i>	RHL1403	GTAGTCAGGCAAAGGCTCGA (reverse)
<i>ttx-1</i> (C-terminal ALFA)	RHL1510	CACCGCGTAATCAATACTGC
<i>ttx-1</i> (C-terminal ALFA)	RHL1511	GACCAACTACTATCCTGCAG
<i>ttx-1</i> (C-terminal ALFA)	RHL1514	AGCATGGAGAACTTGGTCCA (crRNA, exon 18)

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Reviewer #1 (Public review):

Summary:

Mackie and colleagues compare chemosensory preferences between *C. elegans* and *P. pacificus*, and the cellular and molecular mechanisms underlying them. The nematodes have

overlapping and distinct preferences for different salts. Although *P. pacificus* lacks the *lsy-6* miRNA important for establishing asymmetry of the left/right ASE salt sensing neurons in *C. elegans*, the authors find that *P. pacificus* ASE homologs achieve molecular (receptor expression) and functional (calcium response) asymmetry by alternative means. This work contributes an important comparison of how these two nematodes sense salts and highlights that evolution can find different ways to establish asymmetry in small nervous systems to optimize the processing of chemosensory cues in the environment.

Strengths:

The authors use clear and established methods to record the response of neurons to chemosensory cues. They were able to show clearly that ASEL/R are functionally asymmetric in *P. pacificus*, and combined with genetic perturbation establish a role for *che-1*-dependent *gcy-22.3* in the asymmetric response to NH_4Cl .

Weaknesses:

The mechanism of *lsy-6*-independent establishment of ASEL/R asymmetry in *P. pacificus* remains uncharacterized.

Comments on revisions: Looks good - all the best

<https://doi.org/10.7554/eLife.103796.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

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*The authors use clear and established methods to record the response of neurons to chemosensory cues. They were able to show clearly that ASEL/R are functionally asymmetric in *P. pacificus*, and combined with genetic perturbation establish a role for *che-1*-dependent *gcy-22.3* in the asymmetric response to NH_4Cl .*

Weaknesses:

*The mechanism of *lsy-6*-independent establishment of ASEL/R asymmetry in *P. pacificus* remains uncharacterized.*

We thank the reviewer for recognizing the novel contributions of our work in revealing the existence of alternative pathways for establishing neuronal lateral asymmetry without the *lsy-6* miRNA in a divergent nematode species. We are certainly encouraged now to search for genetic factors that alter the exclusive asymmetric expression of *gcy-22.3*.

Reviewer #2 (Public review):

Summary:

*In this manuscript, Mackie et al. investigate gustatory behavior and the neural basis of gustation in the predatory nematode *Pristionchus pacificus*. First, they show that the behavioral preferences of *P. pacificus* for gustatory cues differ from those reported for *C. elegans*. Next, they investigate the molecular mechanisms of salt sensing in *P. pacificus*. They show that although the *C. elegans* transcription factor gene *che-1* is expressed specifically in the ASE neurons, the *P. pacificus* *che-1* gene is expressed in the Ppa-ASE and Ppa-AFD neurons. Moreover, *che-1* plays a less critical role in salt chemotaxis in *P. pacificus* than *C. elegans*. Chemogenetic silencing of Ppa-ASE and Ppa-AFD neurons results in more severe chemotaxis defects. The authors then use calcium imaging to show that both Ppa-ASE and Ppa-AFD neurons respond to salt stimuli. Calcium imaging experiments also reveal that the left and right Ppa-ASE neurons respond differently to salts, despite the fact that *P. pacificus* lacks *lsy-6*, a microRNA that is important for ASE left/right asymmetry in *C. elegans*. Finally, the authors show that the receptor guanylate cyclase gene *Ppa-gcy-23.3* is expressed in the right Ppa-ASE neuron (Ppa-ASER) but not the left Ppa-ASE neuron (Ppa-ASEL) and is required for some of the gustatory responses of Ppa-ASER, further confirming that the Ppa-ASE neurons are asymmetric and suggesting that Ppa-GCY-23.3 is a gustatory receptor. Overall, this work provides insight into the evolution of gustation across nematode species. It illustrates how sensory neuron response properties and molecular mechanisms of cell fate determination can evolve to mediate species-specific behaviors. However, the paper would be greatly strengthened by a direct comparison of calcium responses to gustatory cues in *C. elegans* and *P. pacificus*, since the comparison currently relies entirely on published data for *C. elegans*, where the imaging parameters likely differ. In addition, the conclusions regarding Ppa-AFD neuron function would benefit from additional confirmation of AFD neuron identity. Finally, how prior salt exposure influences gustatory behavior and neural activity in *P. pacificus* is not discussed.*

Strengths:

*(1) This study provides exciting new insights into how gustatory behaviors and mechanisms differ in nematode species with different lifestyles and ecological niches. The results from salt chemotaxis experiments suggest that *P. pacificus* shows distinct gustatory preferences from *C. elegans*. Calcium imaging from Ppa-ASE neurons suggests that the response properties of the ASE neurons differ between the two species. In addition, an analysis of the expression and function of the transcription factor Ppa-*che-1* reveals that mechanisms of ASE cell fate determination differ in *C. elegans* and *P. pacificus*, although the ASE neurons play a critical role in salt sensing in both species. Thus, the authors identify several differences in gustatory system development and function across nematode species.*

*(2) This is the first calcium imaging study of *P. pacificus*, and it offers some of the first insights into the evolution of gustatory neuron function across nematode species.*

(3) This study addresses the mechanisms that lead to left/right asymmetry in nematodes. It reveals that the ASER and ASEL neurons differ in their response properties, but this asymmetry is achieved by molecular mechanisms that are at least partly distinct from

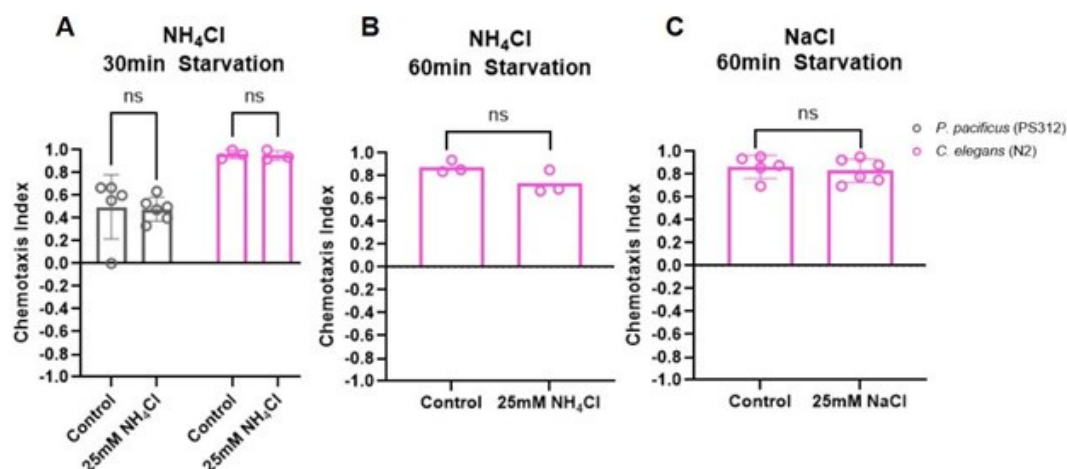
those that operate in *C. elegans*. Notably, ASEL/R asymmetry in *P. pacificus* is achieved despite the lack of a *P. pacificus* *lsy-6* homolog.

Weaknesses:

(1) The authors observe only weak attraction of *C. elegans* to NaCl. These results raise the question of whether the weak attraction observed is the result of the prior salt environment experienced by the worms. More generally, this study does not address how prior exposure to gustatory cues shapes gustatory responses in *P. pacificus*. Is salt sensing in *P. pacificus* subject to the same type of experience-dependent modulation as salt sensing in *C. elegans*?

We tested if starving animals in the presence of a certain salt will result in those animals avoiding it. However, under our experimental conditions we were unable to detect experiencedependent modulation either in *P. pacificus* or in *C. elegans*.

Author response image 1.



(2) A key finding of this paper is that the *Ppa-CHE-1* transcription factor is expressed in the *Ppa*-AFD neurons as well as the *Ppa*-ASE neurons, despite the fact that *Ce-CHE-1* is expressed specifically in *Ce*-ASE. However, additional verification of *Ppa*-AFD neuron identity is required. Based on the image shown in the manuscript, it is difficult to unequivocally identify the second pair of *CHE-1*-positive head neurons as the *Ppa*-AFD neurons. *Ppa*-AFD neuron identity could be verified by confocal imaging of the *CHE-1*-positive neurons, co-expression of *Ppa-che1p::GFP* with a likely AFD reporter, thermotaxis assays with *Ppa-che-1* mutants, and/or calcium imaging from the putative *Ppa*-AFD neurons.

In the revised manuscript, we provide additional and, we believe, conclusive evidence for our correct identification of *Ppa*-AFD neuron being another *CHE-1* expressing neuron. Specifically, we have constructed and characterized 2 independent reporter strains of *Ppa-ttx-1*, a putative homolog of the AFD terminal selector in *C. elegans*. There are two pairs of *ttx-1p::rfp* expressing amphid neurons. The anterior neuronal pair have finger-like endings that are unique for AFD neurons compared to the dendritic endings of the 11 other amphid neuron pairs (no neuron type has a wing morphology in *P. pacificus*). Their cell bodies are detected in the newly tagged *TTX-1::ALFA* strain that co-localize with the anterior pair of *che-1::gfp*-expressing amphid neurons (n=15, J2-Adult).

We note that the identity of the posterior pair of amphid neurons differs between the *ttx-1p::rfp* promoter fusion reporter and TTX-1::ALFA strains– the *ttx-1p::rfp* posterior amphid pair overlaps with the *gcy-22.3p::gfp* reporter (ASER) but the TTX-1::ALFA posterior amphid pair do not overlap with the posterior pair of *che-1::gfp*-expressing amphid neurons (n=15). Given that there are 4 splice forms detected by RNAseq (Transcriptome Assembly Trinity, 2016; www.pristionchus.org), this discrepancy between the *Ppa-ttx-1* promoter fusion reporter and the endogenous expression of the Ppa-TTX-1 C-terminally tagged to the only splice form containing Exon 18 (*ppa_stranded_DN30925_c0_g1_i5*, the most 3' exon) may be due to differential expression of different splice variants in AFD, ASE, and another unidentified amphid neuron types.

Although we also made reporter strains of two putative AFD markers, *Ppa-gcy-8.1* (*PPA24212*)*p::gfp*; *csuEx101* and *Ppa-gcy-8.2* (*PPA41407*)*p::gfp*; *csuEx100*, neither reporter showed neuronal expression.

(3) *Loss of Ppa-che-1 causes a less severe phenotype than loss of Ce-che-1. However, the loss of Ppa-che-1::RFP expression in ASE but not AFD raises the question of whether there might be additional start sites in the Ppa-che-1 gene downstream of the mutation sites. It would be helpful to know whether there are multiple isoforms of Ppa-che-1, and if so, whether the exon with the introduced frameshift is present in all isoforms and results in complete loss of Ppa-CHE-1 protein.*

According to www.pristionchus.org (Transcriptome Assembly Trinity), there is only a single detectable splice form by RNAseq. Once we have a Ppa-AFD-specific marker, we would be able to determine how much of the AFD terminal effector identify (e.g. expression of *gcy-8* paralogs) is effected by the loss of *Ppa-che-1* function.

(4) *The authors show that silencing Ppa-ASE has a dramatic effect on salt chemotaxis behavior. However, these data lack control with histamine-treated wild-type animals, with the result that the phenotype of Ppa-ASE-silenced animals could result from exposure to histamine dihydrochloride. This is an especially important control in the context of salt sensing, where histamine dihydrochloride could alter behavioral responses to other salts.*

We have inadvertently left out this important control. Because the HisCl1 transgene is on a randomly segregating transgene array, we have scored worms with and without the transgene expressing the co-injection marker (*Ppa-egl-20p::rfp*, a marker in the tail) to show that the presence of the transgene is necessary for the histamine-dependent knockdown of NH_4Br attraction. This control is added as Figure S2.

(5) *The calcium imaging data in the paper suggest that the Ppa-ASE and Ce-ASE neurons respond differently to salt solutions. However, to make this point, a direct comparison of calcium responses in C. elegans and P. pacificus using the same calcium indicator is required. By relying on previously published C. elegans data, it is difficult to know how differences in growth conditions or imaging conditions affect ASE responses. In addition, the paper would be strengthened by additional quantitative analysis of the calcium imaging data. For example, the paper states that 25 mM NH_4Cl evokes a greater response in ASE than 250 mM NH_4Cl , but a quantitative comparison of the maximum responses to the two stimuli is not shown.*

We understand that side-by-side comparisons with *C. elegans* using the same calcium indicator would lend more credence to the differences we observed in *P. pacificus* versus published findings in *C. elegans* from the past decades, but are not currently in a position to conduct these experiments in parallel.

(6) It would be helpful to examine, or at least discuss, the other *P. pacificus* paralogs of *Ce-gcy22*. Are they expressed in *Ppa-ASER*? How similar are the different paralogs? Additional discussion of the *Ppa-gcy-22* gene expansion in *P. pacificus* would be especially helpful with respect to understanding the relatively minor phenotype of the *Ppa-gcy-22.3* mutants.

In *P. pacificus*, there are 5 *gcy-22*-like paralogs and 3 *gcy-7*-like paralogs, which together form a subclade that is clearly distinct from the 1-1 *Cel-gcy-22*, *Cel-gcy-5*, and *Cel-gcy-7* orthologs in a phylogenetic tree containing all rGCs in *P. pacificus*, *C. elegans*, and *C. briggsae* (Hong et al, eLife, 2019). In Ortiz et al (2006 and 2009), *Cel-gcy-22* stands out from other ASER-type *gcy* genes (*gcy-1*, *gcy-4*, *gcy-5*) in being located on a separate chromosome (Chr. V) as well as in having a wider range of defects in chemoattraction towards salt ions. Given that the 5 *P. pacificus* *gcy-22*-like paralogs are located on 3 separate chromosomes without clear synteny to their *C. elegans* counterparts, it is likely that the *gcy-22* paralogs emerged from independent and repeated gene duplication events after the separation of these *Caenorhabditis* and *Pristionchus* lineages. Our reporter strains for two other *P. pacificus* *gcy-22*-like paralogs either did not exhibit expression in amphid neurons (*Ppa-gcy-22.1p::GFP*,) or exhibited expression in multiple neuron types in addition to a putative ASE neuron (*Ppa-gcy-22.4p::GFP*). We have expanded the discussion on the other *P. pacificus* *gcy-22* paralogs.

(7) The calcium imaging data from *Ppa-ASE* is quite variable. It would be helpful to discuss this variability. It would also be helpful to clarify how the ASE and ASER neurons are being conclusively identified during calcium imaging.

For each animal, the orientation of the nose and vulva were recorded and used as a guide to determine the ventral and dorsal sides of the worm, and subsequently, the left and right sides of the worm. Accounting for the plane of focus of the neuron pairs as viewed through the microscope, it was then determined whether the imaged neuron was the worm's left or right neuron of each pair. We added this explanation to the Methods.

(8) More information about how the animals were treated prior to calcium imaging would be helpful. In particular, were they exposed to salt solutions prior to imaging? In addition, the animals are in an M9 buffer during imaging - does this affect calcium responses in *Ppa-ASE* and *Ppa-AFD*? More information about salt exposure, and how this affects neuron responses, would be very helpful.

Prior to calcium imaging, animals were picked from their cultivation plates (using an eyelash pick to minimize bacteria transfer) and placed in loading solution (M9 buffer with 0.1% Tween20 and 1.5 mM tetramisole hydrochloride, as indicated in the Method) to immobilize the animals until they were visibly completely immobilized.

(9) In Figure 6, the authors say that *Ppa-gcy-22.3::GFP* expression is absent in the *Ppa-che1(ot5012)* mutant. However, based on the figure, it looks like there is some expression remaining. Is there a residual expression of *Ppa-gcy-22.3::GFP* in ASE or possibly ectopic expression in AFD? Does *Ppa-che-1* regulate rGC expression in AFD? It would be helpful to address the role of *Ppa-che-1* in AFD neuron differentiation.

In Figure 6C, the green signal is autofluorescence in the gut, and there is no GFP expression detected in any of the 55 *che-1(-)* animals we examined. We are currently developing AFDspecific rGC markers (*gcy-8* homologs) to be able to examine the role of *Ppa-CHE-1* in regulating AFD identity.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Abstract: 'how does sensory diversity prevail within this neuronal constraint?' - could be clearer as 'numerical constraint' or 'neuron number constraint'.

We have clarified this passage as '...constraint in neuron number'.

(2) 'Sensory neurons in the *Pristionchus pacificus*' - should get rid of the 'the'.

We have removed the 'the'.

(3) Figure 2: We have had some good results with the ALFA tag using a similar approach (tagging endogenous loci using CRISPR). I'm not sure if it is a *Pristionchus* thing, or if it is a result of our different protocols, but our staining appears stronger with less background. We use an adaptation of the Finney-Ruvkin protocol, which includes MeOH in the primary fixation with PFA, and overcomes the cuticle barrier with some LN2 cracking, DTT, then H₂O₂. No collagenase. If you haven't tested it already it might be worth comparing the next time you have a need for immunostaining.

We appreciate this suggestion. Our staining protocol uses paraformaldehyde fixation. We observed consistent and clear staining in only 4 neurons in CHE-1::ALFA animals but more background signals from TTX-1::ALFA in Figure 2I-J in that could benefit from improved immunostaining protocol.

(4) Page 6: 'By crossing the *che-1* reporter transgene into a *che-1* mutant background (see below), we also found that *che-1* autoregulates its own expression (Figure 2F), as it does in *C. elegans*' - it took me some effort to understand this. It might make it easier for future readers if this is explained more clearly.

We understand this confusion and have changed the wording along with a supporting table with a more detailed account of *che-1p::RFP* expression in both ASE and AFD neurons in wildtype and *che-1(-)* backgrounds in the Results.

(5) Line numbers would make it easier for reviewers to reference the text.

We have added line numbers.

(6) Page 7: is 250mM NH₄Cl an ecologically relevant concentration? When does off-target/nonspecific activation of odorant receptors become an issue? Some discussion of this could help readers assess the relevance of the salt concentrations used.

This is a great question but one that is difficult to reconcile between experimental conditions that often use 2.5M salt as point-source to establish salt gradients versus ecologically relevant concentrations that are very heterogenous in salinity. Efforts to show *C. elegans* can tolerate similar levels of salinity between 0.20-0.30 M without adverse effects have been recorded previously (Hu *et al.*, Analytica Chimica Acta 2015; Mah *et al.* Expedition 2017).

(7) It would be nice for readers to have a short orientation to the ecological relevance of the different salts - e.g. why *Pristionchus* has a particular taste for ammonium salts.

Pristionchus species are entomophilic and most frequently found to be associated with beetles in a necromenic manner. Insect cadavers could thus represent sources of ammonium

in the soil. Additionally, ammonium salts could represent a biological signature of other nematodes that the predatory morphs of *P. pacificus* could interpret as prey. We have added the possible ecological relevance of ammonium salts into the Discussion.

(8) Page 11: 'multiple *P. pacificus* *che-1p::GCaMP* strains did not exhibit sufficient basal fluorescence to allow for image tracking and direct comparison'. 500ms exposure to get enough signal from *RCaMP* is slow, but based on the figures it still seems enough to capture things. If image tracking was the issue, then using *GCaMP6s* with *SL2-RFP* or similar in conjunction with a beam splitter enables tracking when the *GCaMP* signal is low. Might be an option for the future.

These are very helpful suggestions and we hope to eventually develop an improved *che1p::GCaMP* strain for future studies.

(9) Sometimes *C. elegans* genes are referred to as '*C. elegans* [gene name]' and sometimes '*Cel* [gene name]'. Should be consistent. Same with *Pristionchus*.

We have now combed through and corrected the inconsistencies in nomenclature.

(10) Pg 12 - '...supports the likelihood that AFD receives inputs, possibly neuropeptidergic, from other amphid neurons' - the neuropeptidergic part could do with some justification.

Because the AFD neurons are not exposed directly to the environment through the amphid channel like the ASE and other amphid neurons, the calcium responses to salts detected in the AFD likely originate from sensory neurons connected to the AFD. However, because there is no synaptic connection from other amphid neurons to the AFD neurons in *P. pacificus* (unlike in *C. elegans*; Hong et al, eLife, 2019), it is likely that neuropeptides connect other sensory neurons to the AFDs. To avoid unnecessary confusion, we have removed "possibly neuropeptidergic."

(11) Pg16: the link to the Hallam lab codon adaptor has a space in the middle. Also, the paper should be cited along with the web address (Bryant and Hallam, 2021).

We have now added the proper link, plus in-text citation. https://hallemmlab.shinyapps.io/Wild_Worm_Codon_Adapter/ (Bryant and Hallem, 2021)

Full citation:

Astra S Bryant, Elissa A Hallem, The Wild Worm Codon Adapter: a web tool for automated codon adaptation of transgenes for expression in non-*Caenorhabditis* nematodes, *G3 Genes | Genomes | Genetics*, Volume 11, Issue 7, July 2021, jkab146, <https://doi.org/10.1093/g3journal/jkab146>

Reviewer #2 (Recommendations for the authors):

(1) In Figure 1, the legend states that the population tested was "J4/L4 larvae and young adult hermaphrodites," whereas in the main text, the population was described as "adult hermaphrodites." Please clarify which ages were tested.

We have tested J4-Adult stage hermaphrodites and have made the appropriate corrections in the text.

(2) The authors state that "in contrast to *C. elegans*, we find that *P. pacificus* is only moderately and weakly attracted to NaCl and LiCl, respectively." However, this statement

does not reflect the data shown in Figure 1, where there is no significant difference between *C. elegans* and *P. pacificus* - both species show at most weak attraction to NaCl.

Although there is no statistically significant difference in NaCl attraction between *P. pacificus* and *C. elegans*, NaCl attraction in *P. pacificus* is significantly lower than its attraction to all 3 ammonium salts when compared to *C. elegans*. We have rephrased this statement as relative differences in the Results and updated the Figure legend.

(3) In Figure 1, the comparisons between *C. elegans* and *P. pacificus* should be made using a two-way ANOVA rather than multiple *t*-tests. Also, the sample sizes should be stated (so the reader does not need to count the circles) and the error bars should be defined.

We performed the 2-way ANOVA to detect differences between *C. elegans* and *P. pacificus* for the same salt and between salts within each species. We also indicated the sample size on the figure and defined the error bars.

Significance:

For comparisons of different salt responses within the same species:

- For *C. elegans*, NH_4Br vs NH_4Cl (** $p < 0.01$), NH_4Cl vs NH_4I (* $p < 0.05$), and NH_4Cl vs NaCl (* $p < 0.05$). All other comparisons are not significant.

- For *P. pacificus*, all salts showed (**** $p < 0.0001$) when compared to NaAc and to NH_4Ac , except for NH_4Ac and NaAc compared to each other (ns). Also, NH_4Cl showed (* $p < 0.05$) and NH_4I showed (*** $p < 0.001$) when compared with LiCl and NaCl. All other comparisons are not significant.

For comparisons of salt responses between different species (N2 vs PS312):

- NH_4I and LiCl (* $p < 0.05$); NaAc and NH_4Ac (**** $p < 0.0001$)

(4) It might be worth doing a power analysis on the data in Figure 3B. If the data are underpowered, this might explain why there is a difference in NH_4Br response with one of the null mutants but not the other.

For responses to NH_4Cl , since both *che-1* mutants (rather than just one) showed significant difference compared to wildtype, we conducted a power analysis based on the effect size of that difference (~1.2; large). Given this effect size, the sample size for future experiments should be 12 (ANOVA).

For responses to NH_4Br and given the effect size of the difference seen between wildtype (PS312) and ot5012 (~0.8; large), the sample size for future experiments should be 18 (ANOVA) for a power value of 0.8. Therefore, it is possible that the sample size of 12 for the current experiment was too small to detect a possible difference between the ot5013 alleles and wildtype.

(5) It would be helpful to discuss why silencing *Ppa-ASE* might result in a switch from attractive to repulsive responses to some of the tested gustatory cues.

For similar assays using *Ppa-odr-3p::HisCl1*, increasing histamine concentration led to decreasing C.I. for a given odorant (myristate, a *P. pacificus*-specific attractant). It is likely that the amount of histamine treatment for knockdown to zero (i.e. without a valence change) will differ depending on the attractant.

(6) *The statistical tests used in Figure 3 are not stated.*

Figure 3 used Two-way ANOVA with Dunnett's post hoc test. We have now added the test in the figure legend.

(7) *It would be helpful to examine the responses of ASER to the full salt panel in the Ppa-gcy-22.3 vs. wild-type backgrounds.*

We understand that future experiments examining neuron responses to the full salt panel for wildtype and *gcy-22.3* mutants would provide further information about the salts and specific ions associated with the GCY-22.3 receptor. However, we have tested a broader range of salts (although not yet the full panel) for behavioral assays in wildtype vs *gcy-22.3* mutants, which we have included as part of an added Figure 8.

(8) *The controls shown in Figure S1 may not be adequate. Ideally, the same sample size would be used for the control, allowing differences between control worms and experimental worms to be quantified.*

Although we had not conducted an equal number of negative controls using green light without salt stimuli due to resource constraints (6 control vs ~10-19 test), we provided individual recordings with stimuli to show that conditions we interpreted as having responses rarely showed responses resembling the negative controls. Similarly, those we interpreted as having no responses to stimuli mostly resembled the no-stimuli controls (e.g. WT to 25 mM NH_4Cl , *gcy22.3* mutant to 250 mM NH_4Cl).

(9) *An osmolarity control would be helpful for the calcium imaging experiments.*

We acknowledge that future calcium imaging experiments featuring different salt concentrations could benefit from osmolarity controls.

(10) *In Figure S7, more information about the microfluidic chip design is needed.*

The chip design features a U-shaped worm trap to facilitate loading the worm head-first, with a tapered opening to ensure the worm fits snugly and will not slide too far forward during recording. The outer two chip channels hold buffer solution and can be switched open (ON) or closed (OFF) by the Valvebank. The inner two chip channels hold experimental solutions. The inner channel closer to the worm trap holds the control solution, and the inner channel farther from the worm trap holds the stimulant solution.

We have added an image of the chip in Figure S7 and further description in the legend.

(11) *Throughout the manuscript, the discussion of the salt stimuli focuses on the salts more than the ions. More discussion of which ions are eliciting responses (both behavioral and neuronal responses) would be helpful.*

In Figure 7, the *gcy-22.3* defect resulted in a statistically significant reduction in response only towards NH_4Cl but not towards NaCl , which suggests ASER is the primary neuron detecting NH_4^+ ions. To extend the description of the *gcy-22.3* mutant defects to other ions, we have added a Figure 8: chemotaxis on various salt backgrounds. We found only a mild increase in attraction towards NH_4^+ by both *gcy-22.3* mutant alleles, but wild-type in their responses toward Cl^- , Na^+ , or I^- . The switch in the direction of change between the behavioral (enhanced) and calcium imaging result (reduced) suggests the behavioral response to ammonium ions likely involves additional receptors and neurons.

Minor comments:

(1) The full species name of "*C. elegans*" should be written out upon first use.

We have added 'Caenorhabditis elegans' to its first mention.

(2) In the legend of Figure 1, "N2" should not be in italics.

We have made the correction.

(3) The "*che-1*" gene should be in lowercase, even when it is at the start of the sentence.

We have made the correction.

(4) Throughout the manuscript, "HisCl" should be "HisCl1."

We have made these corrections to 'HisCl1'.

(5) Figure 3A would benefit from more context, such as the format seen in Figure 7A. It would also help to have more information in the legend (e.g., blue boxes are exons, etc.).

(6) "Since NH_4I sensation is affected by silencing of *che-1*(+) neurons but is unaffected in *che-1* mutants, ASE differentiation may be more greatly impacted by the silencing of ASE than by the loss of *che-1*": I don't think this is exactly what the authors mean. I would say, "ASE function may be more greatly impacted..."

We have changed 'differentiation' to 'function' in this passage.

(7) In Figure 7F-G, the AFD neurons are referred to as AFD in the figure title but AM12 in the graph. This is confusing.

Thank you for noticing this oversight. We have corrected "AM12" to "AFD".

(8) In Figure 7, the legend suggests that comparisons within the same genotype were analyzed. I do not see these comparisons in the figure. In which cases were comparisons within the same genotype made?

Correct, we performed additional tests between ON and OFF states within the same genotypes (WT and mutant) but did not find significant differences. To avoid unnecessary confusion, we have removed this sentence.

(9) The nomenclature used for the transgenic animals is unconventional. For example, normally the calcium imaging line would be listed as *csuEx93[Ppa-che-1p::optRCaMP]* instead of *Ppache-1p::optRCaMP(csuEx93)*.

We have made these corrections to the nomenclature.

(10) Figure S6 appears to come out of order. Also, it would be nice to have more of a legend for this figure. The format of the figure could also be improved for clarity.

We have corrected Figure S6 (now S8) and added more information to the legend.

(11) Methods section, Chemotaxis assays: "Most assays lasted ~3.5 hours at room temperature in line with the speed of *P. pacificus* without food..." It's not clear what this

| means. Does it take the worms 3.5 hours to crawl across the surface of the plate?

Correct, *P. pacificus* requires 3-4 hours to crawl across the surface of the plate, which is the standard time for chemotaxis assays for some odors and all salts. We have added this clarification to the Methods.

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